Never-Realized Capital Gains

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Appreciated assets are subject to capital gains tax when sold by their original owner. Yet under policies of "stepped-up basis," many countries forgive this latent tax obligation if the asset is instead transferred, unsold, to the owner's heir. In the first part of this paper, we create novel data on capital gains in Norway. A large fraction of household wealth, especially at the top, comes from unrealized asset appreciation and thus contains latent capital gains tax liability. Much of this latent liability is never taxed: outflows from the stock of unrealized capital gains are relatively rare and Norway's stepped-up basis policy gave yearly exemptions equal to 19-25% of the yearly tax base of realized capital gains. In the second part of the paper, we study how realization and intergenerational transfers changed when Norway moved from a system of stepped-up basis to a carryover system in which heirs inherit their predecessor's latent capital gains tax obligation when they inherit appreciated assets. Using two difference-in-difference empirical strategies that exploit cross-sectional variation in either portfolio composition or investor age, we estimate that the removal of step-up precipitated large increases in taxable realizations (a 31% increase in capital gains tax collections) and a corresponding reduction in intergenerational transfers attributable to highly-appreciated stock.

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Introduction

Capital gains are taxed when *realized*: typically, not until the underlying asset is sold.[1] But some capital gains are never realized by their original owner: the asset is instead transferred, unsold, to the original owner's heir. If a policy of "stepped-up basis" is in place for this transfer, the original owner's appreciation will avoid capital gains taxation entirely. Under stepped-up basis, an heir who sells a previously inherited asset will pay capital gains tax only on appreciation during her own holding period.[2] But no one will ever pay capital gains tax on the appreciation that occurred during her predecessor's holding period. Twelve OECD countries, including the US and UK, currently apply stepped-up basis to at least some intergenerational transfers.[3]

Does the opportunity to avoid tax on inherited capital gains depress taxable realizations? If investors' only goal is to avoid capital gains taxation, stepped-up basis provides an opportunity to ensure their gains will never be taxed. A long theoretical literature has therefore argued that taxable realizations would be higher in a world without step-up (Holt and Shelton, 1962; Feldstein and Yitzhaki, 1978; Auerbach, 1989; Kiefer, 1990). But it is not immediately clear how much we should expect investors to respond to a tax break that happens after the wealth has left their possession (and frequently after they themselves are dead and gone). If bequests are motivated primarily by precautionary saving or an egoistic desire for wealth accumulation (as opposed to an altruistic concern for one's heirs), we would not expect any response to such an incentive. Additionally, removing step-up can only have a large effect on realizations if there is a large outstanding stock of unrealized capital gains which the owners expect to eventually bequest, and we do not currently have sufficient

[1]Circumstances other than sales can count as "realization events" (and therefore trigger the capital gains tax) depending on local law and administrative practice. Examples usually include company liquidations and certain types of mergers.

[2]Throughout this paper, we use the terms "inherited assets" and "inherited capital gains" to refer to assets or capital gains received in intergenerational transfers regardless of whether the transfer occurred during the older generation's lifetime or at their death.

[3]The majority of OECD countries that levy both inheritance and capital gains taxes step-up unrealized capital gains at death (OECD, 2021). Specifically: Chile, France, Korea, Lithuania, Portugal, Slovenia, Spain, the United Kingdom, and the United States all apply step-up to most deathtime inheritance transfers; while Denmark, Finland, and Hungary have more complicated regimes in which step-up applies to only some types of inherited assets. Most countries do not apply step-up to capital gains passed as *inter vivos* gifts.

information to assess whether this is so.

Empirical work is therefore needed to determine the magnitude of unrealized and inherited capital gains and the extent to which the tax treatment of inherited capital gains affects taxable realizations. However, two key challenges have inhibited prior empirical study of these questions. The first challenge is that national tax authorities do not produce statistics on unrealized or inherited capital gains. As a result, not much is known about capital gains that may eventually be inherited, or which taxpayers benefit from the stepped-up basis exemption. The second challenge is that, despite many proposals to modify the tax treatment of inherited capital gains, few policy changes have actually been enacted in a way that allows for empirical evaluation of the resulting changes.[4] The United States in 2010, for example, briefly repealed its estate tax and required inheritors of capital assets in testamentary bequests to "carry over" their predecessors' unrealized capital gains for future taxation, but then reverted back to its original system before the end of the year.[5] In this paper, we study the case of the only other country that has substantially modified its treatment of inherited capital gains in the last four decades: Norway.[6]

This paper overcomes the prior empirical challenges by building new data on unrealized capital gains and introducing a previously unstudied reform that removed stepped-up basis in Norway. We combine uniquely rich transaction and wealth data from the Norwegian Tax Authority to estimate a yearly panel of the unrealized capital gains in every Norwegian stock, private business, or real estate holding.[7] We observe whether and when these assets are sold

[4]Presidents Obama and Biden both proposed making death a deemed realization event in their budget proposals to Congress (Treasury, 2024). Alternatively, Biden's 2020 campaign proposed to replace step-up with carryover basis. In the UK, the Institute for Fiscal Studies recently called on the new Labour government to replace the stepped-up basis regime with either carryover basis or deemed realization at death (Adam et al., 2024).

[5]In the end, estates of Americans who died in 2010 were allowed to choose between paying the estate tax and receiving stepped-up basis, or receiving carryover basis treatment. Gordon et al. (2016a,b) evaluate what may be learned from this brief, "voluntary" tax regime. The US also passed legislation that repealed step-up in favor of carryover basis in 1976, but implementation was delayed and the change was eventually repealed before it came into effect. See Gutman (2021) for a discussion of the administrative issues that motivated the repeal of the 1976 reform.

[6]In Section 3, we describe relevant institutional information, including various elements of taxation in Norway.

[7]We cannot estimate unrealized capital gains in mutual funds, but we explore the implications of this lack of coverage in detail in Section 1. International equity is primarily held through mutual funds.

or transferred to heirs, and thus can quantify the amount and distribution of inherited capital gains passing via intergenerational transfers each year.

In the first part of the paper, we establish the importance of unrealized and inherited capital gains in Norway during the stepped-up basis regime. Unrealized capital gains are large and important for understanding top wealth: seventeen to nineteen percent of total household wealth is composed of unrealized capital gains in assets subject to the capital gains tax and that fraction rises above forty percent for the wealthiest households. On a yearly basis, there is very little outflow from the stock of unrealized capital gains into either taxable realization or the stepped-up basis exemption. However, comparing these outflows to each other reveals that the amount of inherited capital gains which step-up exempts from tax is a substantial fraction of the tax base of realized capital gains. Finally, our data reveals that unrealized and inherited capital gains differ from the realized capital gains more commonly observed in tax data. Unrealized and inherited capital gains are predominately from private businesses and are concentrated in the very top of the wealth distribution. In contrast, realized capital gains are predominately from liquid assets and are realized by the somewhat-less-wealthy.

In the second part of the paper, we examine how the removal of step-up in Norway affected taxable realizations and intergenerational transfers. Prior to 2006, Norway allowed capital gains to disappear from the tax base when gifted or bequested (a policy of "stepped-up basis" for both *inter vivos* gifts and testamentary bequests). In 2006, Norway began "carrying over" inherited capital gains to recipients of stock (including private business interests and mutual fund shares) transferred as gifts or bequests. We present a conceptual framework which clarifies that the removal of step-up increases the cost of inheriting appreciated stock. Altruistic investors were differentially exposed to this cost change depending on their age and their portfolio's ratio of unrealized capital gains in stock to total value. Investors with unappreciated stock or who lacked altruistic motives did not experience any change in the cost of transferring their assets to their heirs. Our first empirical strategy exploits the differential exposure to the reform for different portfolios. We estimate difference-in-difference regression models that compare the effect of the reform between investors who have and do not have unrealized capital gains in stock prior to the reform. The key identifying assumption is that, absent the reform, total realizations (of any asset type) would have evolved similarly across investors who accumulated different unrealized capital gains in their stock portfolios. We find that investors whose portfolios were exposed to the removal of step-up increased their yearly taxable realizations by an average of \$2,023 — a 59% increase. Total government revenues from the capital gains tax were 31% higher following step-up's removal than they would have been otherwise. We investigate heterogeneity in responses to the reform and find that the increase in taxable realizations was largest among the investors with the most highly appreciated stock portfolios and investors with children.

As a second, alternative, empirical strategy, we compare realization responses by age group. We estimate difference-in-difference regression models that compare the effect of the reform for individuals aged 30-49 to groups of older individuals. The identifying assumption for this strategy is that, absent the reform, the shape (but not necessarily the level) of the age-log(realizations) profile would be stable over time. We find that investors in older age groups responded to the removal of step-up by increasing their yearly taxable realizations by \$1,365 - \$1,945 (26 to 89% in the 4 years following the reform). The magnitude of the realization response increases monotonically by age group.

Step-up policy is currently subject to debate in both the US and UK. Our results clarify that step-up benefits very wealthy households and that, relative to other policies that would benefit a similar group, step-up creates a particularly negative fiscal externality. Using a "Marginal Value of Public Funds" framework, we estimate that step-up provides only \$0.43 in benefits for each \$1 of foregone tax revenue. Although this level of efficiency loss could be justified for programs explicitly focused on redistribution, step-up is incurring these efficiency losses in order to transfer benefits to wealthy capital gains owners. Furthermore, this efficiency loss is unnecessary: other types of capital gains tax cuts would create *positive* fiscal externalities. We discuss the opportunity for a budget-neutral and efficiency-enhancing reform that would remove step-up and, assuming the policymaker wanted to continue to give benefits to capital gains owners, would use the resulting revenue to simply lower the capital gains tax rate.

The paper proceeds as follows: Section 1 describes our data, including how we create our novel measure of unrealized capital gains. Section 2 uses our novel data to document several new facts about unrealized and inherited capital gains in Norway under the stepped-up basis regime. Section 3 provides background on Norway's policy change from stepped-up basis to a carryover basis regime. Section 4 provides a conceptual framework for understanding how the tax treatment of inherited capital gains (e.g., stepped-up basis) affects investors' realization decisions. Section 5 examines how the removal of step-up affected taxable realizations and the composition of intergenerational transfers. Section 6 discusses policy implications and the generalizability of our results.

1 Data

Our paper uses data and policy variation from Norway, firstly because Norway is the only country that has enacted a permanent change to its stepped-up basis policy in recent decades.[8] Norway is also a uniquely appropriate setting for this project because the Norwegian Tax Authority collects unusually rich data on wealth and transactions, which allow us to build a novel individual-level measure of unrealized capital gains for the entire country. As background, Norway is a small, open economy with a well-educated population, a sizable middle class, and a significant public sector. The Norwegian population is 5.4 million people and consistently ranks among the wealthiest global populations.[9] Levels of wealth inequality are relatively high, and top wealth is concentrated in real estate and private business interests (see Appendix F).

As in many countries, the Norwegian Tax Authority collects data on realized capital gains (a component of taxable income), but does not measure unrealized capital gains. However, we can use components of Norway's wealth and transactions data to build a new measure of unrealized capital gains ourselves. At the individual-asset level, we calculate unrealized capital gains at the end of each year as the current price of the asset minus the original owner's original purchase cost, which is called the owner's "basis" in the asset. That is,

[8] See Appendix Section A for background on stepped-up basis policy from a global perspective.

[9] In 2023, GDP per capita in Norway was \$87,961, versus \$81,695 in the United States (World Bank, 2023).

Unrealized Capital Gains = Current Price - Basis.[10] Below, we introduce the income, wealth and transactions data and describe our measurement of unrealized capital gain in more detail. Throughout this paper, we convert all currency amounts to 2018 USD using yearly average consumer price indices from Statistics Norway and the NOK-USD exchange rate on December 31, 2018.

1.1 Data on Yearly Wealth & Income

We combine several administrative data sources collected by the Norwegian government. We begin with detailed information about each Norwegian resident's yearly income and wealth, captured in tax return data from 1992 to 2018. Much of these data (nearly all components of income and of financial wealth) are third-party reported from employers, banks, and financial intermediaries. The income data include information on (taxable) realized capital gains at the individual-year level.

The Norwegian wealth data are perceived to be high quality, and we are confident that it captures market values well for most components of financial wealth, such as bank deposits, liabilities, and listed securities. Some caveats and data supplementation, however, are necessary when measuring wealth in real estate and unlisted companies. Although the Norwegian tax authority assesses the value of all domestic real property each year, the quality of the assessment methodology prior to a substantial revision in 2010 is thought to be poor, and the 2010 reform did not improve the quality of the assessments for non-residential property. We therefore replace the tax-assessed value of real estate wealth with alternative property-level values computed by Eika et al. (2020). These alternative values use sales prices from the real estate transaction data (described below), and then interpolate and extrapolate those sale values using house price indices in years when that property does not sell. This procedure is validated and discussed in extensive detail in Eika et al. (2020). Unlisted stock are valued for the purposes of the Norwegian tax system by the book value of the firm's underlying assets. This is likely to be a substantial underestimate of market value, and some types of firm assets, such as goodwill and other intangible assets, are not valued at all. We will therefore

[10] Appendix Section A provides additional background on capital gains and realization-based capital gains taxes as general concepts.

replace this measure with an alternative measure of unlisted stock wealth, discussed further below in Section 1.3.

A few categories of wealth are missing. We do not observe wealth in defined contribution pension plans. Defined contribution plans were not permissible in Norway until 2001 and only begin to have market share during our period of study.[11] Although Norwegians are legally required to report wealth abroad, and we observe these reported values, we will not capture international wealth that Norwegians fail to report to the tax authority. It has been established that offshoring forms of tax evasion are substantial among the wealthy citizens of Norway (Alstadsæter et al., 2019a). Our inability to observe this offshore wealth may mean our estimates understate the amount of unrealized capital gains in portfolios at the top of the wealth distribution. On the other hand, since these assets are, by definition, hidden from the Norwegian tax system, we do not expect realizations or inheritance of these assets to affect our estimates of the step-up reform.

1.2 Transactions Data in Real Estate and Stock

Transactions data from the Norwegian Land Register allow us to observe sales and other transfers of real estate. For nearly all properties in Norway, the data contain information on the last transaction prior to 1993, and all subsequent transactions from 1993 to 2018. Detailed information is recorded for each transaction, including the type of transfer (e.g., an arms-length sale or an inheritance transfer).

In addition to the data on real estate transactions, we have transaction data for listed and unlisted stocks[12] over the period 2003 to 2014. These data contain similarly detailed information on the type of transfer, the identity of the buyer or seller, and the transaction price. The data contain information on the last transaction prior to 2003, and all subsequent transactions from 2003 to 2018. The data on transactions in stock have two key limitations: stock transactions are not observed before 2003, and we do not observe transactions in mutual funds. Mutual funds are an important component of many Norwegians' portfolios, with

[11] Even in 2017, the average wealth in individual and private contribution plans was only around $24,000 (Smogeli and Halvorsen, 2017).

[12] Our category of unlisted stock includes all private business interests other than sole proprietorships.

a higher average participation rate than directly held stock (see Appendix Figure A7) and we know from aggregate statistics that mutual funds are responsible for nearly all realized capital gains that are not from directly owned stock or real estate (Statbank Norway, 2007). However, mutual funds are a much smaller fraction of overall wealth for the richest Norwegians (See Appendix Figures A8, A9). Since we observe wealth in mutual funds but cannot calculate their unrealized capital gains, we will bind our estimates of total unrealized capital gains by making the alternative assumptions that (1) Norwegian households do not have any unrealized capital gains in mutual funds and (2) Norwegian households' mutual fund wealth is 100% unrealized capital gains (i.e., all owners of mutual funds have zero basis in their mutual fund holding). In practice, the treatment of mutual funds does not qualitatively affect any of our results.

1.3 A New Measure of Unrealized Capital Gains

To construct our measure of unrealized capital gains, we first convert the transactions data into a ledger, tracking each individual's asset holdings as the individual buys and sells. Our initial measure of an individual's basis in a held asset is the original purchase price which that individual paid for the asset. We calculate unrealized capital gains each year by subtracting the individual's basis in the asset from the contemporary value of the asset at the end of the year.

We ignore unrealized capital gains in asset classes other than Norwegian stock (including unlisted stock), real estate, and mutual funds. Although capital gains from other asset types, including directly held international stock, directly held business assets, artwork, etc., are subject to the capital gains tax, the aggregate amount of wealth in these categories (and thus, the upper bound for the possible amount of unrealized capital gains) is small.[13] See Appendix Figures A8 and A9.

[13]Most Norwegians participate in international equity markets via mutual funds, and this wealth would therefore be included in our mutual fund numbers.

1.3.1 Tracking Basis and Current Holdings

We create a ledger at the individual-asset level. When an individual buys a new asset, we open a ledger for that individual-asset pair. We record the number of shares initially owned in the asset (or fractional ownership share for real estate), and then we update the number of shares held (or fractional ownership held) as the individual makes subsequent transactions in the asset. We account for stock splits, consolidations and mergers when applicable. Appendix Section D validates our ledger against end-of-year shareholder reports in the Norwegian Shareholder Registry (available for limited liability companies since 2004) and finds that our ledger agrees very closely (an exact match for 97% of current shareholders) with the Registry.

We measure an individual's initial basis in an asset as the asset's original purchase cost, or carryover basis if applicable. To properly account for carryover basis, it is necessary to trace shares moving across ledgers in non-taxable transactions. Basis may be carried over between ledgers due to mergers, changes to share classes, or intergenerational transfers after the removal of step-up (The Norwegian Tax Administration, n.d.a). As the individual buys and sells, we increase and decrease his basis under a first-in-first-out assumption, in keeping with the accounting methods required by Norwegian law. For capital gains that are eventually realized, we can validate our measure of basis by using it to calculate realized capital gains (as the recorded value of the sale minus basis), and then comparing that calculation to the realized capital gains reported in each individual's tax return. This match is also good — 97% of individual yearly tax return agree (within rounding error) with our calculation for the amount of realized capital gains in stock (see Appendix D).

1.3.2 Measuring Yearly Value

To measure unrealized capital gains, we require a yearly measure of each asset's market value, even when that asset is not sold. For listed stock, we obtain the stock's end-of-year closing price from Thomson-Reuters Datastream. For unlisted stock, we begin with the book value measure used by the Norwegian government to value each unlisted company for wealth and inheritance tax purposes. When stepped-up basis was in place, this valuation also served as the heir's new basis for unlisted stock received in inheritance. This measure almost certainly undervalues the value of these companies. We therefore create our own measure of unlisted company value by scaling up the company's sales, income, and asset book values with multipliers computed for a sample of listed firms in the same industry, and then applying a 10% liquidity discount to those values. This procedure follows Smith et al. (2023) and is described in greater detail in Appendix E. When a firm's estimated value is negative, we replace that value with zero, in keeping with the limited liability of the firms considered. When unlisted firm value is directly attributable to firm ownership in listed equity, we value that portion of the firm at the listed equity value.

1.3.3 Inheritance Values and Inherited Capital Gains

Until 2014, the (tax-assessed) values of gifts and bequests were reported to the Norwegian tax authority.[14] However, the inheritance data never included information about the amount of unrealized capital gains passing via intergenerational transfers. We construct this measure using the transactions data (described above) for stock and real estate. We observe intergenerational transfers of stock and real estate with information about the particular asset transferred and the quantity (i.e., number of shares) transferred. Since these transactions do not usually include a recorded value, we calculate the amount of capital gains passing in an intergenerational transfer as the value of the transferred asset at the beginning of the year minus the basis associated with the asset.

1.3.4 Data Contribution

As described above, we measure the outstanding stock of unrealized capital gains in Norway at the individual-asset level. We believe we are the first to measure this stock in non-survey data.[15] However, our effort is related to prior work that creates measures of the yearly

[14] Gifts and bequests were not reported if they are below the lowest tax threshold (see Appendix Table A1) or if the estate is administered through the public probate process. Our transaction-based measure of intergenerational transfers of stock and real estate does not suffer from these limitations.

[15] The US Survey of Consumer Finance (SCF) measures unrealized capital gains — see Looney and Moore (2016). Unfortunately, the SCF, by design, excludes the wealthiest US households, preventing a complete characterization of the distribution of unrealized capital gains. The SCF's cross-sectional structure also makes it difficult to use to analyze behavioral changes over time. At the aggregate level in Norway, Andresen (2022) estimates the sum of all unrealized capital gains in unlisted stock.

flow of accruing capital gains. Several recent papers use Norwegian administrative data to directly measure a flow of accruing capital gains (Eika et al., 2020; Fagereng et al., 2024).[16] In the US, prior work has measured capital gains accruals using yearly changes in house price values, rates of return on stocks and changes in the yearly values of wealth reported in the *Forbes 400* (Larrimore et al., 2021; Yagan, 2023). Other related papers allocate flows of firm profits or retained earnings to shareholders to proxy for capital gains accruals (Thoresen et al., 2012; Wolfson et al., 2016; Fairfield and Luis, 2016; Alstadsæter et al., 2023; Aaberge et al., 2024). These papers either do not or cannot link individuals' initial purchase price (i.e., "basis") to contemporary asset holdings, so they cannot measure the total stock of accumulated unrealized gains, only the yealy change in unrealized gains as they accrue. Our measures of basis and unrealized capital gains are necessary to consider the implications of changes to the capital gains tax base within existing realization-based systems.

Prior attempts to estimate the magnitude of capital gains passing in intergenerational transfers have faced notable data limitations. Early work (Steger, 1957; Okun, 1967; Brannon et al., 1967) looked to US estate tax records, which capture the value of wealth transfers subject to the estate tax, but not the amount of capital gains within the transferred wealth. These papers back out a measure of inherited capital gains by assuming that the ratio of capital gains to value is the same for inherited assets and realized assets. We show in Section 2 that this assumption does not hold in practice: assets inherited under step-up are much more highly appreciated than realized assets. More recent attempts (Poterba and Weisbenner, 2001; Office of Tax Analysis, 2014; Avery et al., 2015) estimate inherited capital gains by applying mortality tables to the household-level measure of unrealized capital gains in the US Survey of Consumer Finance: they measure inherited capital gains as equal to unrealized capital gains for households predicted to die in the next year. Office of Tax Analysis (2014) and Gordon et al. (2016a,b) observe the inherited capital gains of US estates that chose carryover treatment in 2010 (the single year in which the US allowed estates to choose between carryover or step-up treatment). Since estates with large amounts of inherited

[16] Eika et al. (2020) compute their measure of households' yearly accrued capital gains using the same transactions data we use in this paper. However, we are the first to use this (or any similar data) to measure basis and unrealized capital gains, or to measure capital gains passing via intergenerational transfers.

capital gains were incentivized to choose step-up treatment, the amount of inherited capital gains they observe in their sample is almost surely an underestimate. A key contribution of our paper is that it is the first to combine comprehensive data on intergenerational transfers with the amount of capital gains in the transferred assets. We are also the first to measure inherited capital gains in a panel setting, which allows us to examine behavioral responses to changes in tax policy.

2 New Descriptive Facts on Unrealized Capital Gains

In this section, we use our data to document new facts about unrealized and inherited capital gains. Because we are interested in learning about the state of unrealized capital gains under step-up (i.e., prior to the Norwegian policy reforms), we will focus on capital gains in Norway as of end-of-year 2004. As will be described in more detail in Section 3, 2004 was the last year before the removal of step-up was announced.

We focus on capital gains in what we call *taxable asset categories* — that is, asset categories that are usually subject to the capital gains tax.[17] Taxable asset categories in our data include listed and unlisted stock, mutual funds, and taxable real estate.[18] Primary residences and certain other forms of real estate, although present in our data, are exempt from capital gains tax if the owner meets certain holding period requirements.[19] These categories of *non-taxable real estate* do not typically benefit from stepped-up basis, and we consider their capital gains separately in the analysis that follows.

Fact 1. *Unrealized capital gains are large: Seventeen to nineteen percent of total household wealth is composed of unrealized capital gains in taxable asset categories.*

See Table 1 Panel (a). At the end of 2004, \$68.2–\$78.5 billion of unrealized capital gains

[17] The use of the term "taxable asset categories" is not meant to imply that the capital gains within these assets will necessarily be subject to the capital gains tax, only that they would be so subject if realized. Capital gains from retained earnings might instead be paid out as dividends, and thus subject to a dividend tax (if any). And, in accordance with the focus of this paper, capital gains passed in intergenerational transfers under step-up are exempted from the capital gains tax base.

[18] Wealth in other possibly taxable asset categories is a very small share of household wealth (see Appendix Figure A8).

[19] Appendix Section B details the holding period requirements for the real estate exemptions from capital gains taxes.

were outstanding in asset categories subject to the capital gains tax: listed and unlisted stock, mutual funds, and taxable real estate. This amount equals 17–19% of aggregate net wealth held by Norwegian households in 2004 ($412 billion).

We present our measures of total unrealized capital gains in taxable assets as a range to acknowledge our uncertainty regarding the amount of unrealized capital gains in mutual funds. We do not observe mutual funds in our transaction data and are therefore unable to measure the amount of unrealized or inherited capital gains in mutual funds. However, we know that an individual's unrealized capital gains in mutual funds cannot exceed his total wealth in mutual funds, which we observe in our wealth data. We can therefore present a range for the value of unrealized gains including mutual funds: the upper bound of this range assumes that all mutual fund holdings are 100% appreciated. The lower bound of this range assumes zero unrealized capital gains in all mutual fund holdings.

Much of household wealth, naturally, is comprised of assets such as cash and primary residences, which are not (or not usually) subject to the capital gains tax. Of the $123.4 billion of wealth in the taxable asset categories of stock, mutual funds, and taxable real estate, 55–64% of this amount was unrealized capital gains. Within these categories, real estate wealth was less appreciated than wealth in listed or unlisted stock. On average, 29% of wealth in taxable real estate was unrealized gains (similar to 28% in non-taxable real estate). Forty-two percent of wealth in listed stocks was composed of unrealized capital gains and 79% of wealth in unlisted stock was unrealized capital gains (Table 1 Panel (a)).

Fact 2. *The majority of potentially-taxable unrealized capital gains are in private (unlisted) businesses.*

Among all assets subject to the capital gains tax, 68-78% of unrealized capital gains are from unlisted shares (Table 1 Panel (a)). $53.2 billion of unrealized capital gains were present in unlisted shares at the end of 2004. In Norway, top wealth is very concentrated in private business holdings (see Appendix Figure A9). So it is perhaps unsurprising that we also find unrealized capital gains to be concentrated in this type of asset. But this compositional effect is amplified by the fact that unlisted shares are much more appreciated on average

(79%) than holdings of listed stock or taxable real estate (which are on average 42% and 29% appreciated, respectively).

The predominance of private business assets in the stock of unrealized capital gains is in marked contrast to their relative rarity among realized gains. Realized capital gains from unlisted stock composed only 7.9% of the total amount of realized capital gains from stock and real estate in 2004 (Table 1 Panel (b)). Unlisted stock was a greater share of inherited capital gains, composing 61.3% of inherited capital gains in stock and real estate in 2004 (Table 1 Panel (c)).

Fact 3. *Unrealized capital gains are concentrated in wealthy households, and wealthy households' portfolios are disproportionately made up of unrealized capital gains in taxable asset categories.*

Figure 1 Panel (a) shows that ownership of taxable unrealized capital gains in 2004 was concentrated among the wealthiest Norwegian households. The bottom 90% of households owned around 30% of the stock of taxable unrealized capital gains in Norway in 2004. The next 9% of households owned 26–27%, and the top 1% of households owned 41–44% of the stock of unrealized capital gains.[20] In contrast, Panel (b) shows that the distribution of unrealized capital gains in real estate categories not subject to the capital gains tax is much more evenly distributed across the wealth distribution.

The importance of unrealized capital gains within household portfolios also varies across the wealth distribution. Figure 1 Panel (c) shows that taxable unrealized capital gains are less than 10% of wealth for the bottom 80% of households in 2004. The ratio of taxable unrealized capital gains to total household wealth increases sharply at the top of the wealth distribution, reaching a value of over 40% for the wealthiest households.

The skewed distribution of taxable unrealized capital gains is mostly driven by differences in household portfolio composition across the wealth distribution. Wealthy households hold large fractions of their wealth in assets subject to the capital gains tax — holdings of unlisted

[20] As before, we present ranges to incorporate alternative assumptions regarding the amount of unrealized capital gains within mutual funds.

stock are particularly important for the wealthiest households. For example, households in the top 0.01% of net wealth in 2004 own 22% of the total wealth in stock and taxable real estate (see Appendix Figure A9). These top households own a similar share of unrealized capital gains in these asset categories. In contrast, less wealthy households own more of their wealth in cash and primary residences, which are not subject to the capital gains tax. Therefore, because the fraction of wealth in primary residences decreases across the wealth distribution, Figure 1 Panel (d) shows that the fraction of household wealth attributable to unrealized capital gains in non-taxable assets is decreasing in wealth.

Fact 4. *Outflows from the stock of unrealized capital gain are relatively rare. But the outflow to inheritance (and thus to tax exemption under Norway's stepped-up basis policy) is 19-25% of the size of the yearly outflow to taxable realization.*

Yearly outflows of capital gains to either taxable realization or inheritance are very small relative to the stock of unrealized capital gains. $310 million of capital gains in stock and taxable real estate were passed through inheritance and thus exempted from tax under the stepped-up basis policy in 2004. $1.65 billion capital gains were realized and taxed, from a total stock of $68.3 billion of unrealized capital gains (Table 1 column 2). Therefore, in 2004, inherited capital gains in stock and taxable real estate equaled 0.4% of the stock of unrealized capital gains in those categories and realized capital gains equaled 2.4% of the same stock.

Yet, comparing the two (small) outflows to each other reveals one measure of step-up's significance: Since the $1.65 billion of realized capital gains constitutes the base of the capital gains tax in 2004 for stock and real estate, we have shown that Norway's stepped-up basis policy in 2004 exempted a amount equal to 19% of their capital gains tax base. Numbers in 2003 were similar — $274 million of inherited capital gains is equivalent to 25% of that year's amount of realized capital gains from stock and taxable real estate ($1.06 billion).

The fact that only five times are many capital gains were realized as inherited in 2004 is despite the fact that 38 times more value was realized than inherited (Table 1, Panels (b) and (c), column 1). Inherited assets are much more highly appreciated than those that are realized. The ratio of (taxable) capital gains to value in inherited assets subject to the capital gains tax is 57%. Realized assets are much less appreciated: The ratio of capital gains to value in realized assets subject to the capital gains tax is 8%. Listed and unlisted stock show a particularly large difference in appreciation between inherited and realized assets: 65% versus 1% in listed stock, and 71% versus 7% in unlisted. Real estate, whether taxable or non-taxable, does not show a similar pattern (Table 1, Panels (b) and (c), column 3).

Fact 5. *Inheritances that benefit from the stepped-up basis exemption come from disproportionately wealthy households.*

Although Fact 3 (Figure 1) established that unrealized capital gains are disproportionately concentrated at the top of the wealth distribution, wealthy households own an even larger share of capital gains that are passed in intergenerational transfers. Figure 2 shows that the bottom 90% of households are the source of only 14% of inherited capital gains in taxable assets in 2004. The next 9% of households are responsible for 20% of inherited capital gains, while the top 1% is the source of the remaining 66%. The top 0.01% alone is responsible for 34% of inherited capital gains in taxable assets.

The households that transfer large amounts of capital gains in inheritance are the mechanical beneficiaries of stepped-up basis policy: the capital gains they transfer are exempted from the capital gains tax under step-up. Figure 2, therefore, paints an portrait of step-up as a tax break that goes to households that are wealthy even relative to the skewed distribution of capital gains ownership (Figure 1). In the next part of the paper, we explore how investor behavior changes when the stepped-up basis exemption is removed.

3 Norway's Removal of Step-Up for Stock (2005-2006)

The remainder of this paper studies the removal of stepped-up basis for stock in Norway (including listed and unlisted stock, and equity-based mutual fund shares). This change was announced in 2005 and implemented in 2006. Capital gains and inheritance tax rates did not change during the same period. Further detail on later policy changes, including the removal of step-up for non-stock asset classes (concurrent with the removal of the inheritance tax in

2014), can be found in Appendix B.

During our period of study, capital gains realized by their original owner were taxed at a 28% rate.[21] Heirs were subject to an inheritance tax on the market value of received gifts and bequests (of any type), but, until 2006, were not subject to capital gains tax on capital gains from their predecessor's holding period. The top inheritance tax rate for transfers between close relatives was 20% from 1999 to 2008.[22] We end our baseline difference-in-difference results in 2008 to isolate the removal of step-up for stock from the subsequent 2009 change in the inheritance tax rates.

The 2006 removal of step-up for stock was passed as an afterthought to a wider reform of capital taxation (see Appendix Section B.1.4 for more detail). Although the main components of the 2006 tax reform were formally announced in March 2004 and approved in June of that year, the removal of step-up was not part of the June legislation and was not definitively decided until later in the year (Norwegian Ministry of Finance, 2004). Since we do not find any press coverage regarding the impending removal of step-up until early 2005, we take 2005 to be the relevant announcement year when considering anticipatory responses in our empirical analysis.[23]

The removal of step-up for stock meant that, from January 1, 2006, intergenerational transfers of appreciated stock (including unlisted stock and equity-based mutual funds) re-

[21] The capital gains tax rate was 28% from 1992 to 2013 and has never depended on individual income (i.e., there is no progressive rate structure). The same rate was also applied to other forms of capital income, such as interest income. Capital losses can be used to fully offset capital gains. Net losses can be used to offset ordinary income at a disadvantaged rate: $1 of capital losses offsets only $0.72 of ordinary income, and cannot be applied against social security contributions or top marginal tax rates.

[22] Norway's inheritance tax had a rate structure that depended on the cumulative amount of the inheritance received, and the relationship between the giver and the recipient. Other than a small yearly gift exemption, no distinction was made between *inter vivos* gifts and bequests at the time of death. Appendix Table A1 provides details on the rates and brackets over time.

[23] The removal of step-up for stock was formally introduced on December 10, 2004 (Stortinget, 2004). However, this change did not receive immediate media attention. A search through the Norwegian media library (Nasjonalbiblioteket) for the terms "kontinuitetsprinsippet" and "diskontinuitet" (which are the Norwegian terms corresponding to carryover and stepped-up basis, respectively) shows that the first mention of the impending change appeared in the newspaper *Finansavisen* on January 8, 2005, under the headline "One Year Left" (Amundsen, 2005). A second mention appeared in *Levanger-Avisa* on February 19, 2005 (Vigdal, 2005). Both articles featured lawyers advocating that Norwegians transfer stock (especially stock of family businesses) to the younger generation before the new carryover basis regime came into effect. Media coverage of the upcoming change grew more intense during the spring and summer of 2005.

ceived carryover basis treatment.[24] That is, heirs who inherited appreciated stock continued to be subject to inheritance tax on the value of the stock they received, but now were also subject, if and when they sold the stock, to capital gains tax on the amount of appreciation during the original owner's holding period.[25] Losses, however, were not permitted to be carried over in the 2006 reform, so stocks with losses continued to be subject to a "step-down" in basis at the time of inheritance. Step-up also continued for non-stock assets (such as real estate) until 2014.[26]

Our empirical analysis focuses on how stock owners reacted to the 2005 announcement of step-up's impending removal, rather than the later implementation of that removal. The announcement date is important because the capital gains tax treatment of the original owner is not changing. The original owner faces the same cost of realizing his capital gains before or after step-up is removed. But the announcement of step-up's removal informs the original owner that the inheritance he may leave in the future will be more expensive, which will cause him to reassess his trade-off between present realizations and future intergenerational transfers. We formalize this understanding of step-up's effects on owners of unrealized gains in the next section of the paper.

Although repealing stepped-up basis and inheritance taxes are frequently considered in tandem, our empirical setting allows us to isolate the effects of removing step-up from any changes to the inheritance tax regime. Conceptually, an inheritance tax is not a perfect substitute for exempting inherited capital gains from the capital gains tax base. Inheritance transfer taxes apply to the market value of the bequest, regardless of any unrealized

[24]In Norway, mutual fund investors are not taxed when fund managers buy and sell stocks or reinvest dividends. Capital gains taxes are only triggered when the investor sells their mutual fund units. Mutual funds are classified as stock funds and taxed identically to stock if more than 80% of the fund is in equities; otherwise, they are treated as combination funds, with stock income taxed at a higher rate (currently 38%) and bond income taxed at 22%. Exemptions that are available to offset capital gains from stock are also applicable to capital gains from stock in mutual funds.

[25]From 2006 until the removal of the inheritance tax in 2014, recipients of gifted or inherited stock were given a deduction on their inheritance tax bill equal to 20% of the carried-over capital gains tax liability, to lessen the "double taxation" from being subject to both an inheritance tax and carried over capital gains. We abstract from this deduction in our explanation of the policy changes, but fully account for it in all our figures and empirical analyses.

[26]See Appendix B for detail on the removal of step-up for non-stock assets, which is not the focus of our empirical analysis.

capital gains. Under stepped-up basis, an investor who sells a highly appreciated asset the day before death and leaves the resulting cash as inheritance is subject to both the capital gains tax and the inheritance tax, whereas a dying investor who bequests the unrealized asset itself pays only the inheritance tax (if any). As our conceptual framework in Section 4 demonstrates, three distinct policy levers affect the trade-off between capital gains realizations and intergenerational transfers: the capital gains tax rate, the inheritance tax rate, and step-up.[27]

4 Conceptual Framework

This section presents a conceptual framework to understand the effects of step-up. Step-up makes it less costly for heirs to realize inherited assets. Under the assumption that givers of intergenerational transfers have (partially) altruistic bequest motives (i.e., that they care about their heirs' after-tax value of the transfer), we show that step-up will disincentivize realizations by the original owners of appreciated assets.

4.1 A Model of Capital Gains and Bequests

Consider an individual with wealth of value W, some fraction γ of which is unrealized capital gains. The individual may either choose to realize this wealth, in which case he uses the proceeds of that realization for his own current consumption C, or the individual may pass the unrealized wealth to his heirs, who will realize it for their own consumption. The individual chooses after-tax consumption (C) and bequests (B), subject to a budget constraint that depends on two costs: his own cost of realization (c_R) and his heir's cost of inheriting and then realizing (c_{I}):

$$\max_{C,B} \qquad u(C) + A \cdot V(B) \quad \textit{subject to} \qquad W = \frac{C}{(1-c_R)} + \frac{B}{(1-\mu c_{\mathrm{I}})}$$

[27] There is a small existing literature on the interactions between inheritance and capital gains taxation. Prior work illustrates two of the three "policy levers" mentioned above: Auten and Joulfaian (2001) show that higher estate tax rates encourage more capital gains realizations by the original owner. Poterba (2001) provides evidence that larger unrealized capital gains push owners towards the utilization of stepped-up basis. We are the first paper to show empirically that changing stepped-up basis affects taxable realizations.

where A is the individual's utility weight on bequests and $\mu \in [0, 1]$ weights how much the individual cares about his heir's taxes. That is, μ determines the individual's bequest motive. If $\mu = 1$, the individual can be said to have a purely altruistic bequest motive: he treats taxes that his heir pays on the bequest as if he faced those taxes himself. If $\mu = 0$, the individual has a wealth-in-utility or a warm-glow bequest motive and does not care at all about any taxes his heir may face on her inheritance. (He values the wealth he has left over to give rather than how much of a transfer the heir receives.)[28] With a general $\mu \in [0, 1]$, our model nests these two alternative bequest motives and allows for mixed motives between the two extremes.[29]

4.2 The Cost of Realizing and Inheriting

We define a "per-dollar cost of realization" and a "per-dollar cost of inheritance" to illustrate how the capital gains tax, inheritance tax, and tax treatment of inherited capital gains together affect the trade-off between consumption and bequests.

We first define a "per-dollar cost of realization" as the cost to extract 1 unit of value from wealth with fraction γ of unrealized capital gains:

$$c_R \equiv c_R(\gamma) = \tau_c \cdot \gamma$$

where τ_c is the statutory capital gains tax rate and γ is the ratio of capital gains to total value in the realized wealth. All realizations face the same tax rate (τ_c) on realized capital gains. But the fact that the capital gains tax is levied on capital gains, rather than total value, means that c_R will vary across realized assets, because it is cheaper to extract the same amount of value from less-appreciated assets.

As an illustration, suppose two individuals own different stocks with the same current

[28] Givers might care about the pre-tax value of their bequests if they care more about being known to the world as a wealthy and generous individual than about how their bequests are valued by the recipients. In our setting, inheritance taxes are imposed on the heir, which makes wealth-in-utility and warm-glow bequest motives for the giver equivalent. They would differ in a setting in which an estate tax was imposed on the giver. Additionally, our focus is on rational models, but we could also consider $\mu < 1$ to parameterize the giver's inattention/perception of non-salience regarding his heirs' taxes, á la Chetty et al. (2009).

[29] As a model of "impure altruism", our model builds on a tradition from Andreoni (1989).

value. Suppose Stock 1 was originally purchased for \$99 and is now worth \$100 (Unrealized Capital Gains = \$1). Suppose Stock 2 is also currently worth \$100 but was originally purchased for only \$50 (Unrealized Capital Gains = \$50). If the first individual realizes the full \$100 value from Stock 1, it will cost $\tau_c \times \$1$. If the second individual realizes the full \$100 value from Stock 2, it will cost $\tau_c \times \$50$. In per-dollar terms, the cost of realization is thus τ_c times the amount of unrealized capital gains divided by the total value, precisely as shown in the equation for c_R above.

The "per-dollar cost of inheritance" is the heirs' cost of inheriting \$1 of wealth and then immediately realizing the accompanying capital gains (if any).[30]

$$c_I \equiv c_I(\gamma) = \begin{cases} \tau_I & \textit{under stepped-up basis} \\ \tau_I + \tau_c\gamma & \textit{under carryover basis} \end{cases}$$

where τ_I is the statutory inheritance tax rate, τ_c is the statutory capital gains tax rate, and γ is the fraction of capital gains to total value in the inherited wealth. For notational convenience, we will also define the policy indicator θ to equal 0 when a step-up regime is in place and equal 1 when in a carryover basis regime is in place. We can then write the per-dollar cost of inheritance as $c_I = \tau_I + \theta\tau_c\gamma$.

Figure 3 depicts how the per-dollar cost of inheritance, c_I, is affected by the removal of stepped-up basis, depending on the fraction γ of unrealized capital gains to total value. The cost of inheritance c_I increases when step-up is removed, and the magnitude of the change is increasing in the ratio of unrealized capital gains to value (γ). In a stepped-up basis regime, the heir did not have to pay capital gains tax on inherited capital gains. The cost of inheritance was therefore equal to the inheritance tax rate τ_I (the flat dashed line in Figure 3 Panel (a)). When step-up is removed, the heir now has to pay capital gains taxes when he realizes the inherited asset. The total cost of inheritance is thus the sum of the inheritance tax and the heir's capital gains tax obligation, and depends sharply on

[30] Alternatively, if we assume that the heir realizes the inherited wealth T periods after the inheritance is received, we will need to modify the formula for the carryover basis case to account for the distance in time between the payment of inheritance and capital gains taxes: $c_I = \tau_I + \beta^T \cdot \tau_c \cdot \gamma$

the fraction of unrealized capital gains in the asset (the solid light blue line in Figure 3 Panel (a)). In contrast, the cost of realizations for the original owner's consumption, c_R is unaffected by the removal of step-up.

4.3 Effects of Step-Up Policy on Realizations

Returning to the consumption model of Section 4.1, we can rewrite the budget constraint by plugging in our definitions for costs c_R and c_I:

$$W = \frac{C}{1-c_R} + \frac{B}{1-\mu c_I} = \frac{C}{1-\tau_c\gamma} + \frac{B}{1-\mu(\tau_I+\theta\tau_c\gamma)}$$

To maximize utility subject to this budget constraint, our individual will, naturally, choose consumption and bequests such that the ratio of his marginal utilities is set equal to the relative price: $u_B/u_C = (1-c_R)/(1-\mu c_I)$. Changes to step-up policy (i.e., $\Delta\theta$), which affect the per-dollar cost of inheritance c_I, will therefore give rise to income and substitution responses, discussed further below:

Lemma 1 (Altruism is Required for Givers to Care About Step-up)**.** *If $\mu = 0$ (i.e., the giver does not care about his heir's taxes), the removal of step-up will not affect his consumption and bequest decisions.*

Since changes to step-up ($\Delta\theta$) only affect c_I, and c_I only appears in the individual's decision problem when multiplied by μ, changes to c_I do not matter for individuals with $\mu = 0$. There is a large literature on various bequest motives (see Kopczuk (2013) for a review) and no single bequest motive seems to be sufficient to explain observed patterns of wealth and intergenerational transfers. This lemma makes clear that we require some degree of altruism to explain our main empirical findings in Section 5 (where we show large realization responses to the removal of step-up). But we will revisit non-altruistic motives in Section 5.2 in order to explain our observed pattern of anticipation responses.

Proposition 1 (Step-Up Depresses Realizations)**.** *If $\mu > 0$ and the substitution effect dominates the income effect, then replacing stepped-up basis with carryover basis (i.e., a change* *from $\theta = 0$ to $\theta = 1$) will weakly increase the amount C realized for present consumption.*

See Appendix C for formal proof. Briefly: the change in θ weakly raises the cost of bequests relative to present consumption ($\frac{c_I}{c_R} \uparrow$). If the substitution effect dominates the income effect, this price change causes consumers to reallocate their wealth toward current consumption and away from bequests, leading to a net increase in C.

Corollary 1 (Differential Exposure by γ)**.** *Given the conditions in Proposition 1, the magnitude of the increase in realizations for present consumption (C) that follows the removal of step-up ($\Delta\theta = 1$) will depend on the ratio γ of unrealized gains to value within an individual's portfolio. In particular:*

a. *Individuals whose portfolios have some amount of unrealized capital gains ($\gamma > 0$) will strictly increase the amount C realized for present consumption.*

b. *Individuals whose portfolios do not have any unrealized capital gains ($\gamma = 0$) will not change their realization behavior.*

c. *Individuals whose portfolios have relatively larger fractions γ of unrealized capital gains will have relatively larger increases in realizations.*

Since θ only enters the budget constraint in the individual's maximization problem when multiplied by γ, this corollary follows directly from Proposition 1. In our analysis of the removal of step-up for stock in Norway in Section 5, Corollary 1 guides our first empirical strategy. We consider individuals with some amount of unrealized capital gains in stock to be "treated" by the removal of step-up for stock, while individuals without any unrealized gains in stock are "untreated." We also explore heterogeneous treatment effects by γ and confirm that individuals with the most-appreciated stock portfolios have the largest responses to the reform.

Corollary 2 (Differential Exposure by A)**.** *Given the conditions in Proposition 1, individuals with larger utility weights A on bequests will respond to the removal of step-up ($\Delta\theta = 1$) with relatively larger increases in realizations for present consumption (C).*

This corollary follows directly from Proposition 1. In our analysis of the removal of step-up for stock in Norway in Section 5, Corollary 2 guides our age-based empirical strategy. We assume that older individuals and individuals with children place more utility weight on bequests and therefore consider them to be more exposed to the removal of step-up for stock. We confirm that older individuals and people with children have the largest responses to the reform.

In the next section, we begin our empirical analysis of the removal of step-up, guided by the framework and the propositions above.

5 Observed Responses to the Removal of Step-Up

In 2005, following thirteen years without substantial changes to capital gains and inheritance tax rates,[31] Norway announced that step-up would be removed for stock (including private business interests and equity-based mutual funds). Stock gifted or bequested after January 1, 2006, would be subject to capital gains tax on the original owner's gains when the heir next realized the asset.

In this section, we document responses to the 2005-2006 removal of step-up for stock.[32] First, we focus on realization responses and document large increases in taxable realizations. We show that this increase was driven by individuals whose portfolio composition, family structure, and age meant that they were the most exposed to the removal of step-up. Second, we document that this increase in realizations occurred despite the opportunity, throughout 2005 to make early bequests that would still benefit from stepped-up basis treatment. Third, we examine how increased realizations and anticipatory behavior affected aggregate trends for intergenerational transfers.

[31] See Appendix Table A1 for detail on minor modifications to the inheritance tax brackets during this time.

[32] Although step-up was eventually removed for non-stock asset classes in 2014 (see Appendix B), evaluation of that removal is confounded by the 2014 removal of the inheritance tax, which had a simultaneous but opposite effect on the relative price of bequests versus realizations. We therefore choose to focus our analysis only on the earlier reform, which removed step-up for stock while holding inheritance tax rates fixed.

5.1 Capital Gains Realization Responses

For investors with altruistic bequest motives, the removal of step-up made it relatively cheaper to realize appreciated stock relative to the outside option of leaving that stock to their heirs (Section 4.2). Our conceptual framework demonstrated that, in a very simple model of realizations and bequests, the removal of step-up should trigger an increase in realizations, particularly for altruistic investors who are older and whose portfolios have large fractions of unrealized capital gains in stock. In this section, we seek to learn whether the removal of step-up actually caused such an increase in taxable realizations.

5.1.1 Aggregate Increase in Realizations

As an initial naive examination, we look to the aggregate amount of net capital gains realized each year. Capital gains realizations surged following the announcement of the removal of step-up — aggregate realized capital gains more than tripled in 2005 compared to the previous year, and remained high until 2008's financial crisis. Figure 4 documents this change, which corresponded to a 205% increase in tax revenue from the capital gains tax between 2004 and 2005. However, as evidenced by the other fluctuations on display in Figure 4, aggregate capital gains realizations frequently exhibit large year-to-year variation. Large losses tend to be realized during economic downturns, and more gains are realized in periods of economic boom. We are therefore unwilling to attribute the 2004-to-2005 increase in realizations to the removal of step-up without accounting for other time-specific factors such as contemporaneous macroeconomic shocks.

5.1.2 Empirical Strategy using Differential Portfolio Exposure

Not all stock owners were equally exposed to the removal of step-up in 2006. As explained in our conceptual framework (Section 4, Corollary 1), individuals are more or less affected by the inheritance cost shock from the removal of step-up for stock depending on the ratio of unrealized capital gains in stock to value (γ) in their portfolio. The intensity of the realizations response to the removal of step-up should be mechanically increasing for $\gamma \in (0, 1]$, and we do not expect any realization response for individuals with $\gamma \leq 0$ (no unrealized capital gains in stock).[33] The average effect[34] of the removal of step-up on taxable realizations in year t for individuals with a fixed level of exposure γ can be written:

$$ATT_t(\gamma) = E[Y_{it}^{\gamma} - Y_{it}^{0} | \Gamma_i = \gamma]$$

where Y_{it}^{γ} denotes the potential outcome (taxable realizations) of an individual i if she experiences treatment exposure of magnitude γ, and Y_{it}^{0} denotes the potential outcome if the individual is not exposed to the removal of step-up. $\Gamma_i = \gamma$ denotes individual i's actual exposure level γ (because the individual's portfolio had a ratio γ of unrealized capital gains in stock to value in the year before removal).

Although these γ-specific treatment effects reveal important heterogeneity in responses to the removal of step-up, we are primarily interested in the overall effect on total realizations and thus total tax revenue. The average effect of the removal of step-up for all individuals affected by the removal is:

$$ATT_t = E\big[Y_{it} - Y_{it}^{0} | \Gamma_i > 0\big]$$

where Y_{it} represents individual i's observed amount of taxable realizations in year t. This overall ATT_t is a weighted average over all the $ATT_t(\gamma)$ with $\gamma > 0$, and can be transformed into a measure of the total increase in taxable realization engendered by the removal of step-up by simply multiplying the ATT_t by the number of individuals with $\Gamma_i > 0$.

5.1.2.1 Comparing Realization Responses for Differently Exposed Individuals

To account for the concern about confounding time effects, we require a control group that is not exposed to the removal of step-up but whose observed response to macroeconomic shocks and other time effects will allow us to construct a counterfactual for the realization behavior of individuals exposed to step up's removal. Guided by Corollary 1, we use individuals with

[33] Loss positions were not exposed to the 2006 reform because, as discussed in Section 3, the 2006 reform removed step-up for stock but not step-down. That is to say, a carryover basis regime was implemented for intergenerational transfers of appreciated stock, but capital losses were not carried over until the subsequent reform in 2014.

[34] We seek to document the extent to which the removal of step-up increased capital gains realizations for those individuals who were exposed to the removal of step-up. Our target parameter is thus an average treatment effect for the treated (an "ATT").

$\gamma \leq 0$ (negative or zero unrealized capital gains in stock in the year before the removal of step-up) as our control group. Our first empirical strategy is therefore the following: We compare the average taxable realizations of shareholders before and after the removal of step-up, contrasting those individuals who had a positive amount of unrealized capital gains in their stock portfolio (i.e., the treatment group) and individuals with no or negative unrealized capital gains in their stock portfolio (i.e., the control group). To be included in the analysis, an individual must own stock at the beginning of 2005. We then sort these individuals into the treatment or control group depending on whether they have positive appreciation ($\gamma > 0$) in their stock portfolios at the beginning of 2005. Table 2 Panel (a) reports the average change in the per-dollar cost of inheritance for this treatment and control group before and after the 2006 change. Our outcome variable is the individual's net realized capital gains, as reported on his or her yearly income tax return, which includes realized capital gains from stock and from all other taxable asset categories. Table 2 Panel (b) reports the averages of this outcome variable for the treatment and control group. Secondarily, we separate the treatment group into groups based on treatment exposure: we group individuals based on their portfolios' fraction of unrealized capital gains in stock.

These comparisons are only useful if observed realizations by the control group are informative of what realizations by the treatment group(s) would have been if step-up had not been removed. To identify our treatment effects, we assume that realizations for these groups would have trended proportionally. That is, we assume that, if step-up had not been removed, the groups' average realizations would experience the same percent changes over time. Assumption 1 describes this assumption formally:

Assumption 1 (Parallel Trends in Percent Changes)**.** *The average outcome for the group of individuals with treatment exposure γ would have, in the absence of treatment, evolved proportionally to the average outcome of the untreated group:*[35]

[35] An assumption of parallel trends in percent changes is commonly made by assuming parallel trends in a logged variable — but zeros in our outcome variable make a log transformation particularly inappropriate to our setting (Chen and Roth, 2024). Furthermore, estimating a diff-in-diff using OLS on a log transformation of the dependent variable only approximates the proportional difference in growth rates between treated and untreated groups. Such approximations ignore Jensen's inequality and thus may deliver biased estimates of the true multiplicative effect (Ciani and Fisher, 2018). Our estimator avoids this bias and can be estimated with Poisson Pseudo Maximum Likelihood (PPML) regression. PPML consistently estimates

$$\frac{E[Y_{it}^0|\Gamma_i{=}\gamma,Post_t{=}1]-E[Y_{it}^0|\Gamma_i{=}\gamma,Post_t{=}0]}{E[Y_{it}^0|\Gamma_i{=}\gamma,Post_t{=}0]}=\frac{E[Y_{it}^0|\Gamma_i{=}0,Post_t{=}1]-E[Y_{it}^0|\Gamma_i{=}0,Post_t{=}0]}{E[Y_{it}^0|\Gamma_i{=}0,Post_t{=}0]}$$

where $Post_t$ is an indicator that equals one if the year is greater than or equal to 2005 (the announcement year).

We also require a "no anticipation" assumption for the years we consider to be our pre-period:

Assumption 2 (No Anticipation Before 2005)**.**

$$E[Y_{it}^0|\Gamma_i = \gamma, Post_t = 0] = E[Y_{it}^\gamma|\Gamma_i = \gamma, Post_t = 0] \quad \forall t, \gamma$$

Under Assumptions 1 and 2, we can employ a difference-in-difference estimator to identify an $ATT_t(\gamma)$ for each level of treatment exposure γ, from which we can aggregate to our summary parameter of interest for all treated individuals:

$$ATT_t = E[ATT_t(\gamma)|\Gamma_i > 0]$$

We are primarily interested in the ATT_t in levels, because it translates readily into the aggregate increase in capital gains tax revenue. However, we can also identify the effect of the removal of step-up in terms of percent changes from the counterfactual.[36]

5.1.2.2 Adding Covariates Our baseline specification, as laid out above, does not include covariates. However, our identification strategy may be more plausible if we modify our parallel trends assumption (Assumption 1) so that it only needs to hold after conditioning on covariates. For example, under the assumptions that (1) individuals realize to consume in accordance with the permanent income hypothesis and (2) wealth shocks affect portfolios proportionally to their ex-ante capital gains, we can microfound parallel trends in percent changes for individuals of the same age. We therefore present results that condition on indi-

the multiplicative effect as long as the mean function is correctly specified as multiplicative (Wooldridge, 2023).

[36] For the case without covariates, we can also manually compute the treatment effect in percent changes using the summary statistics in Table 2 Panel (b) as (1.25-0.42)/(1+0.42) = 0.58.

viduals' age and features of their stock portfolio. To include these time-invariant covariates (which we will call X_i), we need to modify our two previous assumptions to depend on X_i:

Assumption 1′ (Conditional Parallel Trends in Percent Changes)**.**

$$\frac{E[Y_{it}^0|X_i=x,\Gamma_i=\gamma,Post_t=1]-E[Y_{it}^0|X_i=x,\Gamma_i=\gamma,Post_t=0]}{E[Y_{it}^0|X_i=x,\Gamma_i=\gamma,Post_t=0]} = \frac{E[Y_{it}^0|X_i=x,\Gamma_i=0,Post_t=1]-E[Y_{it}^0|X_i=x,\Gamma_i=0,Post_t=0]}{E[Y_{it}^0|X_i=x,\Gamma_i=0,Post_t=0]}$$

Assumption 2′ (Conditional No Anticipation Before 2005)**.**

$$E[Y_{it}^0|X_i = x, \Gamma_i = \gamma, Post_t = 0] = E[Y_{it}^\gamma|X_i = x, \Gamma_i = \gamma, Post_t = 0] \quad \forall \gamma, t$$

and then add an overlap condition:

Assumption 3 (Overlap)**.**

$$P(\Gamma_i \leq 0|X_i = x) > 0 \quad \forall x$$

The overlap assumption implies that a covariate match can be found for all persons with $\Gamma_i > 0$. If there are regions where the support of X does not overlap for $\Gamma_i \leq 0$ and $\Gamma_i > 0$, then we can only estimate a treatment effect for participants whose covariates lie within the common support region. Under Assumptions 1′, 2′ and 3, we can identify a treatment effect for each (γ, x) pair, and aggregate to $ATT_t = E[ATT_t(\gamma, x)|\Gamma_i > 0]$.

5.1.2.3 Estimation We recover the ATT_t from the parameters of the following regressions using Poisson Quasi-Maximum Likelihood (implemented using Correia et al. (2020)). In the case without covariates, we estimate the regression:

$$Y_{it} = exp\Big(\beta_0 + (Treat_i \times Post_t)\beta_1 + Treat_i\beta_2 + Post_t\beta_3\Big)\epsilon_{it} \tag{1}$$

with $E[\epsilon_{it}|Treat_i, Post_t] = 1$. In our main analysis we compare all treated individuals to the untreated control group, and so we define $Treat_i \equiv \mathbb{1}\{\Gamma_i > 0\}$. In our analyses that instead break out treatment effects by the intensity of treatment exposure, we run regressions that each compare a bin of treatment intensity (e.g. $\gamma \in [\gamma^{low}, \gamma^{high}]$) to the control group ($\gamma \leq 0$). For each of these analyses, we define treatment as, e.g., $Treat_i \equiv \mathbb{1}\{\Gamma_i \in [\gamma^{low}, \gamma^{high}]\}$. The exponentiated coefficient $\exp(\beta_1) - 1$ can be interpreted as the percentage change in realizations due to the policy change and all the coefficients together allow us to construct the treatment effect in levels:

$$ATT = exp(\beta_0 + \beta_1 + \beta_2 + \beta_3) - exp(\beta_0 + \beta_2 + \beta_3) \tag{2}$$

In the case with saturated, time-invariant covariates, we can estimate the following regression:

$$\begin{gathered} Y_{it} = \\ exp\Big(\beta_0 + (Treat_i \times Post_t)\beta_1 + Treat_i\beta_2 + Post_t\beta_3 + \\ X_i\beta_4 + (Treat_i \times X_i)\beta_5 + (Post_t \times X_i)\beta_6 + \\ (Treat_i \times Post_t \times \dot{X}_i)\beta_7\Big)\epsilon_{it} \end{gathered} \tag{3}$$

Where $\dot{X}_i \equiv X_i - E[X_i|\Gamma_i = \gamma]$ denotes that the covariates in the final interaction have been demeaned. Again, $\exp(\beta_1) - 1$ has a percentage-change interpretation, and the parameters allow us to recover the ATT:

$$ATT(x) = exp(\beta_0 + \beta_1 + \beta_2 + \beta_3 + x(\beta_4 + \beta_5 + \beta_6) + \dot{x}\beta_7) - exp(\beta_0 + \beta_2 + \beta_3 + x(\beta_4 + \beta_5 + \beta_6))$$

We also estimate year-by-year event study estimates by replacing the $Post_t$ variable in the equations above with an indicator variable for each specific year.

5.1.2.4 Results Column 1 of Table 3 reports our baseline estimates: individuals exposed to the removal of step-up increased their taxable capital gains realizations by $2,023 (2018 USD) per year. This is a very large behavioral response — in terms of percentage change, these individuals' realizations are 59% higher than their counterfactual level. Figure 5 displays year-by-year event study estimates of the ATT. These year-by-year estimates allow us to visually examine trends in the variable prior to 2005 and the time pattern of the response after 2005. The pattern (in terms of percent changes) of average realizations for the treatment and control group appear similar in the pre-period. A large increase in realizations for the treatment group began as soon as the repeal of step-up was announced (2005), and higher levels of realizations in the treatment group are sustained throughout the post-period.

Given the one-time nature of the removal of step-up, it is not possible to fully separate transitory and permanent responses to the removal of step-up in our data. Theoretically, we expect a short-run increase in realizations as investors rearrange their portfolios in response to the change, but also (as predicted by Proposition 1) a sustained higher level of realizations in the new equilibrium. Our main difference-in-difference results (Table 3) concentrate on the years immediately following the removal of step-up, and these results might therefore combine both short-run and long-run increases. However, when we look to the year-by-year responses (Figure 5) we see sustained higher levels of realizations even thirteen years after step-up's removal.[37] Furthermore, we believe that the combined (short-run and long-run) response is policy relevant: countries are currently considering changes to step-up policy because they are interested in keeping the current stock of unrealized capital gains in the tax base, not only because it will change the accumulation of capital gains in the future.[38]

As shown at the bottom of Table 3, our ATT estimates can be converted into estimates of increased tax revenues simply by multiplying the ATT by the number of treated individuals and the capital gains tax rate. Since 344,538 individuals were in the treated group, our estimates imply that the removal of step-up triggered \$697 million of additional capital gains realizations, implying \$195.16 million of increased tax revenue (relative to the counterfactual). This implies that 24% of the (notably high) level of collected capital gains tax revenue in 2005 was attributable to the removal of step-up and that the removal of step-up

[37]The early years (2005-2008) appear to display slightly larger effects in Figure 5, but this difference is not statistically significant, and the later year coefficients may also be biased downwards due to inheritance tax rate cuts in 2009 and 2014.

[38]The short-run response may be particularly interesting due to generational differences in wealth accumulation. Baby boomers and the silent generation (individuals born between 1928 and 1964) in western countries have benefited from a historic run-up in asset prices during their adult lifetime. Some public debate has focused on a need to reform inheritance taxation prior to the "great wealth transfer" expected as these generations begin to die in larger numbers (Smith, May 14, 2023).

can be said to have increased capital gains tax revenue 31% above counterfactual in 2005.

Our baseline results consider all treated individuals as a group, but Figure 6 shows that responses to the removal of step-up were heterogeneous depending on individuals' level of exposure. The size of the average response increases monotonically in the degree of exposure to the reform. We find no evidence of statistically-significant responses among investors with low levels of appreciation in their stock portfolios. Results appear to be driven by extremely large responses among the most-appreciated investors. Results are also driven by individuals who are parents. Table 4 shows that the response for non-parents is less than half the magnitude of the overall response and does not reach conventional levels of statistical signficance.

Table 3 also examines how our estimates are affected by the inclusion of time-invariant covariates. These covariates are included by saturating the regression with indicator variables for each covariate level. In column (2), we add an indicator for whether an individual owns any unlisted stock (as is common at the top of the Norwegian wealth distribution). In column (3) we further add indicators for the decade bin of an individual's age. As these covariates are added, our estimates grow slightly in magnitude but remain qualitatively very similar.

5.1.3 Alternative Empirical Strategy using Age

As an alternative empirical strategy, we compare realization responses for different age groups, using a group of younger stock owners as a control group. This approach requires different assumptions than our first empirical strategy, but produces qualitatively similar results.

5.1.3.1 Age comparisons in the raw data We begin by graphing the age-log(realizations) profile in Figure 7, normalizing by comparing each age's average realizations to the average realizations of 30-to-40 year-olds in the same year. In the pre-period (from 1994 to 2004), the shape of this profile appears to be fairly stable across years. Averaging across pre-period years (as in Panel (c)), realizations peaked around age 50 at a level approximately 27% higher than those of 30-to-40 year-olds, and then decreased for older ages, so that average realizations at age 70 were approximately 4% below the realizations of 30-to-40 year-olds. Figure 7 also provides initial evidence that this relationship changes in the initial years after the announcement of step-up's removal in 2005: realizations peaked at age 50 at a level approximately 47% higher than those of 30-to-40 year-olds and declined slowly, so that realizations by 70-year-olds were 40% higher those of 30-to-40 year-olds in 2005-2008.

5.1.3.2 Diff-in-Diff Identification To formalize these comparisons, we set up a difference-in-difference empirical strategy using younger age groups (specifically, 30-49 year olds) as our control group. Define A_{it} as individual i's age group in year t, normalized (for notational convenience) so that $A_{it} = 0$ indicates our control group (30-49 year olds). We can then write Y_{it}^a to denote the potential outcome (taxable realizations) of an individual i if she was aged $A_{it} = a$ in year t, and Y_{it}^0 as the potential outcome if individual i is aged $A_{it} = 0$ (i.e., 30 to 49) in year t.

To identify an average treatment effect for a given age group (an $ATT_t(a) \equiv E[Y_{it}^a - Y_{it}^0 | A_{it} = a]$), we need to assume that, if step-up had not been removed, the older groups' counterfactual realizations would experience the same percent changes as the younger control group's realizations. Assumption 1″ formalizes this assumption:

Assumption 1″ (Parallel Trends in Percent Changes, by Age)**.**

$$\frac{E[Y_{it}^0 | A_{it} = a, Post_t = 1] - E[Y_{it}^0 | A_{it} = a, Post_t = 0]}{E[Y_{it}^0 | A_{it} = a, Post_t = 0]} =$$

$$\frac{E[Y_{it}^0 | A_{it} = 0, Post_t = 1] - E[Y_{it}^0 | A_{it} = 0, Post_t = 0]}{E[Y_{it}^0 | A_{it} = 0, Post_t = 0]}$$

In other words, in the absence of the policy change, yearly shocks may move the age-log(realizations) profile up and down, but the curvature of the profile (i.e., the relative realization level of different ages) is assumed to be stable over time. Figure 7 Panel(a) illustrates that the curvature of this profile was indeed fairly stable during our pre-period.

As before, we also require a no anticipation assumption:

Assumption 2″ (No Anticipation Before 2005, by Age).

$$E[Y_{it}^0|A_{it} = a, Post_t = 0] = E[Y_{it}^\gamma|A_{it} = a, Post_t = 0] \quad \forall a, t$$

5.1.3.3 Estimation We estimate difference-in-difference regressions that use individuals aged 30-49 years as the control group. We compare this group, in turn, to individuals aged 50-59, 60-69, 70-79, and 80-89 years. For consistency with our baseline analysis, we only include individuals who own some stock at the beginning of 2005. For each comparison, we estimate the regression:

$$Y_{it} = exp\Bigg(\beta_0 + (Old_{it} \times Post_t)\beta_1 + Old_{it}\beta_2 + Post_t\beta_3\Bigg)\epsilon_{it} \tag{4}$$

with $E[\epsilon_{it}|Old_{it}, Post_t] = 1$, where $Old_{it} \equiv \mathbb{1}\{Age_{it} > 0\}$ is an indicator for being in the older included age group. As before, the exponentiated coefficient $\exp(\beta_1) - 1$ can be interpreted as the percentage change in realizations for the treated group due to the policy change, and all the coefficients together (as in Equation 2) allow us to construct an average treatment effect for the treated (older) age group.

5.1.3.4 Results Table 5 presents the estimated responses by age group. We estimate treatment effects in percent changes that are monotonically increasing in age, although the level response is largest for the 70-79 age group due to a higher baseline. The level increase in taxable capital gains realizations ranges from $\$1,365$ (for individuals aged 50-59) to $\$1,945$ (for individuals aged 70-79). Interestingly, despite the different empirical strategy, the estimated treatment effects are fairly similar to our baseline estimated increase ($2,023) in Table 3.

5.2 Anticipatory Bequest Responses

We find large realization responses to the removal of step-up despite the fact that Norwegians had a full year (all of 2005) in which to make anticipatory inheritance transfers that would still fall under the stepped-up basis regime. As discussed in Section 3, legislation to remove step-up for stock was passed on December 10, 2004, but only implemented for inheritance transfers beginning on January 1, 2006. In the interim period, Norwegians were able to make *inter vivos* transfers that would still benefit from stepped-up basis. Several news articles during 2005 urged Norwegians, especially private business owners, to consider early inheritance transfers (Amundsen, 2005; Vigdal, 2005). Consistent with this advice, an unprecedented number of stock-based gift transactions took place in the later months of 2005. In particular, Figure 8 documents that inheritance transfers in late 2005 contained an outlier amount of capital gains in stock.

These anticipatory responses have two important implications for interpreting our realization results in Section 5.1. First, without these anticipatory responses, we would expect weakly larger (short-run) realization responses than what we document in Section 5.2.[39] Second, the fact that we find realization responses beginning in the announcement year, 2005, (Figure 5) means that many Norwegians deliberately passed up the opportunity to benefit from step-up by making anticipatory transfers. The fact that we see an increase in immediate realizations reveals that early inheritance transfers were an imperfect substitute for the previously available option of benefiting from step-up via later intergenerational transfers.

Reasoning why early inheritance transfers may be costly relative to later transfers returns us to the question of bequest motives. A fully altruistic giver (e.g., one with $\mu = 1$ in our conceptual framework in Section 4), if he knew exactly how much stock he planned to leave in inheritance, should respond to the impending removal of step-up by making an earlier-than-planned intergenerational transfer. Ignoring this opportunity despite the tax consequences suggests that some portion of the bequest motive must either come from precautionary saving or a wealth-in-utility motive. The giver must want to maintain lifetime control over the planned bequest, either due to uncertainty over his own consumption needs, or because he values control of his stockpiled wealth directly.[40] Yet some portion of the bequest motive

[39] The anticipatory responses were only possible because of the lag between the announcement and implementation of step-up's removal and the fact that Norway allowed step-up for *inter vivos* gifts.

[40] This finding parallels a stylized fact from prior work on gift taxation, in which givers appear to forgo a large amount of tax savings in order to keep their wealth in their own possession as long as possible. That is, although *inter vivos* gifts do increase in response to tax incentives (Joulfaian, 2004; Ohlsson, 2011), they do not respond as much as they would if the giver chose the timing of his transfer (lifetime vs deathtime) with only tax considerations in mind (Kopczuk, 2007, 2013).

must also be altruistic (i.e., valuing the post-tax amount received by the heir) else there should be no behavioral response to the removal of step-up. All together, our results imply that individuals value the post-tax amounts received by their heirs, but are willing to forgo some tax benefits to maintain longer control over their wealth.

5.3 Aggregate Inherited Capital Gains in the Post-Period

Above, we documented an increase in realizations of highly-appreciated stock by the original owners. This should presumably imply less capital gains in stock transferred in inheritance after 2006. Indeed, although the total value of all forms of inheritance has grown in every year for which there is data, the composition of inheritance appears to shift away from stock and perhaps away from highly appreciated stock in particular following the 2006 reform. Figure 9 Panel (a) shows that the value of inherited stock is flat over time, even as the value of all types of inheritance continues to grow. Figure 9 Panel (b) shows that inherited stock had a somewhat smaller fraction of unrealized capital gains to value after 2006. Together, these two panels imply that the ratio of capital gains in stock to total inheritance value has fallen over time.

6 Interpretation

We have shown in this paper that stepped-up basis, relative to carryover basis, involves a substantial expenditure of government funds and that the benefits of step-up accrued to the wealthiest households in the Norwegian economy. Step-up policy costs the government money both through the mechanical effect of exempting inherited capital gains and by depressing overall taxable realizations. Continuing a policy of step-up, then, should be thought of as a deliberate fiscal policy choice: it involves a cost (both directly and in terms of distortionary effects), and it provides a benefit to a small number of individuals (mechanically, the heirs who receive and realize inherited capital gains, and indirectly, the original owners of those inherited assets). How should the government think about step-up as a deliberate fiscal policy? In this section, we examine the efficacy of step-up compared to other possible policies. We demonstrate that even if policymakers wish to maintain the same level of tax benefits for capital gains owners, step-up is an unusually cost-ineffective way to give such a benefit. The MVPF quantifies the "leakiness" of Okun's famous bucket metaphor for redistributive government policies: "The money must be carried from the rich to the poor in a leaky bucket. Some of it will simply disappear in transit, so the poor will not receive all the money that is taken from the rich" (Okun, 1975). Lower MVPFs imply that, as the particular benefit is transferred to the beneficiaries, more government revenue than usual leaks out due to behavioral responses. We calculate an MVPF of 0.43, which implies that, accounting for behavioral distortions, each \$0.43 of tax benefits from step-up (given to those who inherit appreciated assets) costs the government \$1 of foregone tax revenue.

6.1 Welfare Implications for Norway's Removal of Step-up

We consider the implications of our results for social welfare in a marginal value of public funds (MVPF) framework (Wildasin, 1984; Mayshar, 1990; Hendren, 2016):

$$MVPF = \frac{1}{1 + FE}$$

where FE is the effect on government revenue from behavioral responses (i.e., the "fiscal externality") per dollar of (mechanical) government expenditure on the program.

Using the simplifying assumptions of our model in Section 4, the (negative) fiscal externality from a marginal change in step-up policy θ is the increase in capital gains tax revenue from behavioral responses, minus the decreased inheritance tax revenue from those same responses:[41]

$$FE_{d\theta}(\theta) = -\big(\tau_c(1-\theta) - \frac{\tau_I}{\gamma}\big)\frac{dRCG/d\theta}{\tau_c ICG(\theta)} \tag{5}$$

where γ is the appreciation ratio, τ_I and τ_c are inheritance and capital gains tax rates, $dRCG/d\theta$ is the realizations response and $ICG(\theta)$ is the amount of inherited capital gains under policy θ (so the mechanical expenditure on step-up is $\tau_c ICG(\theta)$). However, we empha-

[41]The negative sign is necessary because a marginal increase in θ (i.e., less step-up) corresponds to a decrease in direct government expenditure. This formula can be calculated from our model in Section 4 as the implied change to the government's budget constraint from a change in θ, divided by the heirs' willingness to pay to avoid that change, minus 1.

size that Equation 5 is the formula for a marginal policy change and employing this first-order approach to empirical welfare analysis from starting point $\theta = 0$, although common in the literature, will overstate the fiscal externality of the discrete policy change we evaluate.[42] We therefore follow Kleven (2021)'s suggestion to assume linearity between endpoints, and approximate the fiscal externality from the full removal of step-up as:

$$FE \approx -\frac{1}{2}\left(\frac{1}{ICG(0)} - \frac{\tau_I}{\tau_c \gamma}\left(ICG(0)^{-1} + ICG(1)^{-1}\right)\right)\Delta RCG$$

where we measure γ as the average appreciation ratio of treated assets before the policy change, ΔRCG as the reduced-form realization response ($=N_{TREAT} \times ATT$) from Section 5, and $ICG(\theta)$ as the observed levels of inherited capital gains under step-up and carryover, adjusting for growth in wealth over time.With this approach, we calculate a FE of 1.33 and an MVPF of .43.

Comparing our MVPF estimate with estimates of MVPFs for other government programs (Hendren and Sprung-Keyser, 2020) reveals step-up to be associated with a larger (more negative) fiscal externality than other types of tax policy. An MVPF of 0.43 is only slightly smaller than standard for spending programs targeted at adults, in which the government trades off a desire to redistribute money or other goods with the program's distortionary effect on labor income. Compared with other tax breaks, however, the negative fiscal externality of step-up stands out. MVPF estimates for tax rate cuts usually exceed 1, which would imply a positive fiscal externality: each dollar of tax rate cuts costs the government less than a dollar of foregone revenue.[43] This is because cutting the tax rate reduces distortions from the tax system, which encourages (taxable) economic activity. Step-up, in contrast, reduces tax collections from inheritors of appreciated assets (who notably come from extremely wealthy backgrounds, see Section 2) while increasing overall distortions (i.e., creating negative fiscal externalities) to the capital gains tax system.

[42] Note that a marginal change from a carryover baseline (i.e., $\theta = 1$, where there is fully neutral capital gains treatment between the older and younger generation) has no fiscal externality from the capital gains tax, only from the changes to inheritance tax revenue.

[43] The MVPF for a reduction in the capital gains tax rate is $\frac{1}{1+\epsilon_\tau}$, where ϵ_τ is the (usually negative) elasticity of taxable capital gains realizations with respect to the capital gains tax rate.

If the government wished to give a tax benefit to owners of capital gains in a more cost-effective manner, repealing step-up could be paired with a reduction in capital gains tax rates. Such a policy change would be distortion-reducing, benefit a similar group of individuals,[44] and could be accomplished in a budget-neutral manner.[45] Given the existence of this budget-neutral reform and the fact that step-up's beneficiaries are so disproportionately wealthy, it is hard to justify the continuation of step-up on either equity or efficiency grounds.

6.2 Implications for Other Countries

Although our data and policy changes take place in Norway, we believe our findings are relevant to ongoing policy debates about the taxation of inherited capital gains in other countries. We find a large increase in taxable realizations when step-up is removed, which persists throughout our post-period, even after the inheritance tax is fully repealed in Norway. This finding contrasts with the US Joint Committee on Taxation's model of step-up removal, which concludes that the magnitude of increased realizations expected from removing step-up would not be as large as the expected decrease in realizations from removing the estate tax (Joint Committee on Taxation, 2012). Although we expect the effects of the removal of step-up to differ across institutional contexts, we believe policymakers should update their priors in the direction of large behavioral effects.

Because Norway granted step-up to *inter-vivos* transfers, the removal of step-up in Norway featured a year-long period during which owners of capital gains could make early inheritance transfers that would still benefit from step-up. The availability of this anticipatory response means that we expect smaller short-run realization responses in Norway, relative to what we would expect for the removal of step-up in other countries which grant step-up only for deathtime bequests. There is also arguably less incentive to defer capital gains in Norway,

[44]Although both policies would benefit capital gains owners, the beneficiaries of a tax rate cut would be anyone who realizes capital gains, while step-up gives tax benefits to those who realize inherited capital gains specifically. In our data (see Appendix Figure A3), the former population is larger and somewhat less wealthy, which could be seen as an additional reason to favor rate cuts over step-up.

[45]The exact rate reduction that could be financed by the repeal of step-up depends on the elasticity of capital gains realizations with respect to the tax rate *within a carryover basis regime*. We are not aware of existing estimates within the Norwegian context but expect the elasticity to be lower than it was during the stepped-up basis regime.

relative to countries such as the US and UK. This smaller incentive is due to the fact that inter-corporate dividends are fully tax exempt, which allows Norwegians to build up holding companies in which they can reallocate their portfolios without triggering the capital gains tax. Furthermore, Norway paired its stepped-up basis regime with a broad-based inheritance tax, which imposed a cost that needed to be paid to benefit from step-up. In contrast, the estate tax regimes currently in place in the US and UK exempt many inheritances from transfer tax while still granting them stepped-up basis. On the other hand, Norway's yearly wealth tax has been shown to encourage higher levels of savings, particularly in listed stocks and mutual funds (Ring, 2024). It is hard to know exactly how to adjust our estimates for these institutional factors, but the main point is that there is nothing obvious about our empirical setting that would lead us to expect smaller effects elsewhere. Our hope is that future research will be able to study the removal of step-up in other countries, and therefore directly test the external validity of our findings. In the meantime, the conclusion of our policy comparison in Section 6.1 applies as long as our empirical results are *directionally* true. We have shown that step-up is a tax break that amplifies behavioral distortions. Reducing tax rates, in contrast, gives a tax break while simultaneously reducing tax-induced behavioral distortions. It is therefore hard to justify the continuation of step-up policies, even if the goal is to provide a tax break to certain capital-gains-rich individuals.

7 Conclusion

Under policies of "stepped-up basis," many countries exempt inherited assets from previously accumulated capital gains tax liability. A lack of data and policy variation has limited prior empirical work on the distribution and distortionary consequences of this type of policy. To overcome those challenges, this paper builds novel data on unrealized capital gains in Norway. We use this data to document several empirical facts about unrealized and inherited capital gains in a stepped-up basis regime. We show that a large share of average and, particularly, top household wealth is unrealized capital gains, and that the amount of capital gains exempted from tax under step-up is a sizable fraction of the total capital gains tax base.

The second part of this paper presents the first empirical estimates of how the tax treatment of inherited capital gains affects inheritance and taxable capital gains realizations. Stepped-up basis distorts investor behavior, both in terms of reduced taxable capital gains realizations and an altered composition of assets transferred in inheritance. Removing step-up corrects this distortion, dampens the distortionary effects of the capital gains tax overall, and frees up public funds from a tax exemption that disproportionately benefits the wealthiest people in the economy.

Table 1: Unrealized, Realized, and Inherited Capital Gains in 2004 (billions of 2018 USD)

(a) Asset Value & Unrealized Capital Gains

Asset Type	Total Value	UCG	UCG/Value
Taxable Assets including Mutual Funds	123.4	78.48†	0.64†
Mutual Funds	10.2	-	-
Taxable Assets excluding Mutual Funds	113.2	68.28	0.60
Taxable Real Estate	32.4	9.24	0.29
Listed Shares	13.8	5.84	0.42
Unlisted Shares	67.0	53.20	0.79
Nontaxable Real Estate	164.0	46.40	0.28

(b) Realizations & Realized Capital Gains

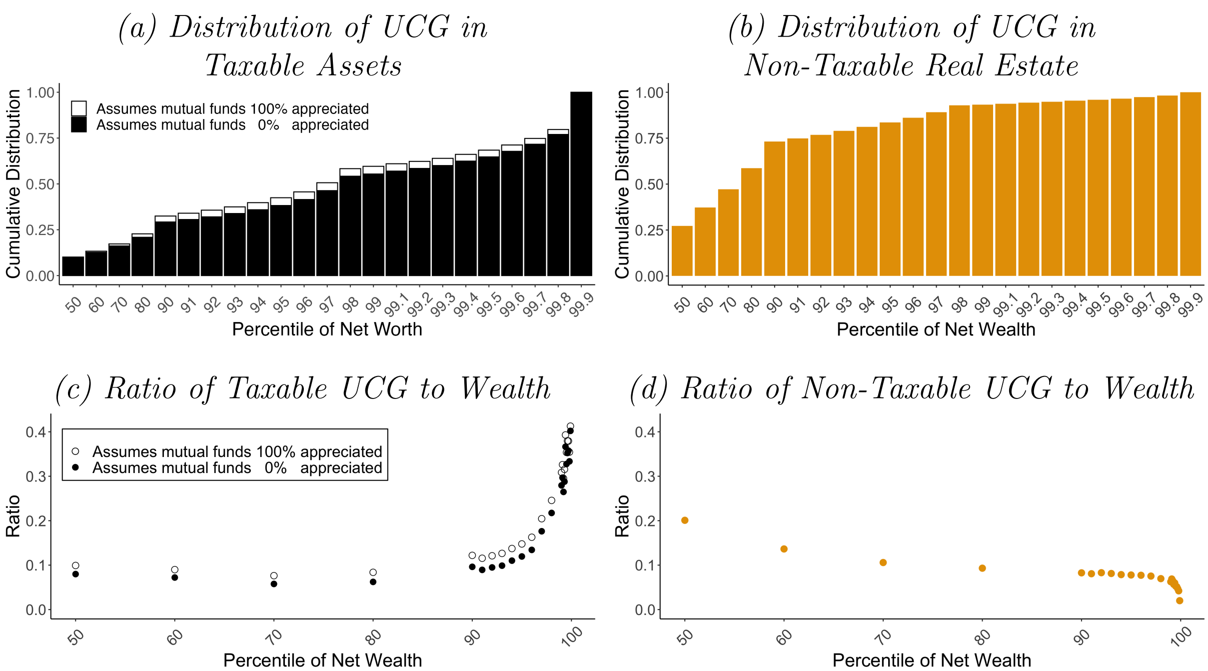
Asset Type	Amount Realized	RCG	RCG/Amount
Taxable Assets excluding Mutual Funds	20.42	1.65	0.08
Taxable Real Estate	2.31	1.29	0.56
Listed Shares	16.20	0.22	0.01
Unlisted Shares	1.91	0.13	0.07
Nontaxable Real Estate	6.29	2.90	0.46

(c) Inheritances & Inherited Capital Gains

Asset Type	Amount Inherited	ICG	ICG/Amount
Taxable Assets excluding Mutual Funds	0.54	0.31	0.57
Taxable Real Estate	0.17	0.05	0.30
Listed Shares	0.10	0.07	0.65
Unlisted Shares	0.27	0.19	0.71
Nontaxable Real Estate	0.51	0.16	0.31

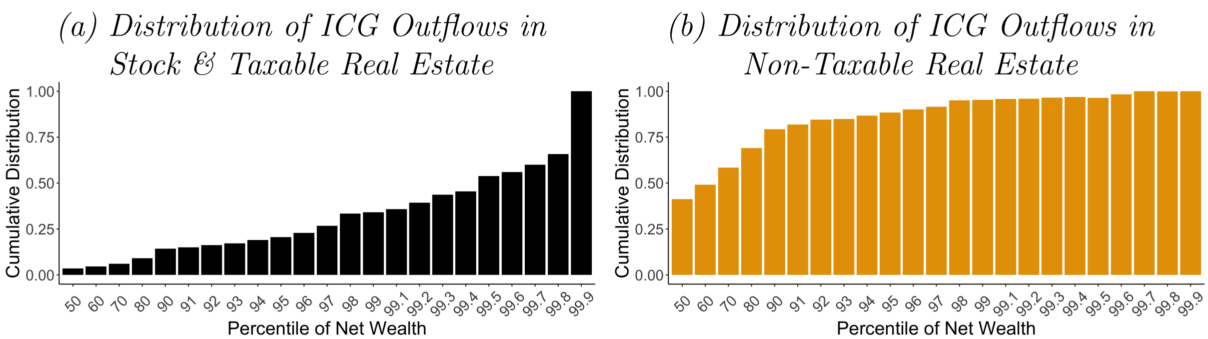
Note: This table presents aggregate numbers on the value of wealth and capital gains in certain asset categories in Norway at the end of 2004. Panel (a) presents the total amount of wealth ("Total Value") in each category, along with the amount of Unrealized Capital Gains ("UCG"). The amount of UCG when mutual funds are included assumes that mutual funds are 100% appreciated (estimates affected by this assumption are marked with a dagger †). Panel (b) displays the value of assets that are realized in 2004 ("Amount Realized"), along with the Realized Capital Gains ("RCG"). Panel (c) displays the value of assets that are inherited in 2004 ("Amount Inherited"), along with the Inherited Capital Gains ("ICG").

Figure 1: Unrealized Capital Gains Across the Wealth Distribution, 2004



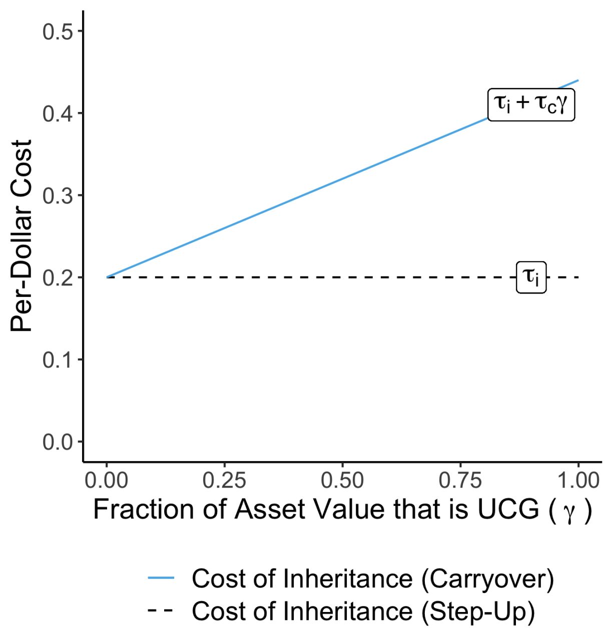
Note: This figure displays the cumulative distribution of unrealized capital gains and the average ratio of unrealized capital gains to total net wealth, for household net wealth percentiles in 2004. Panel (a) depicts the distribution of unrealized capital gains in assets subject to the capital gains tax (mutual funds, listed and unlisted stock and taxable real estate). Because we do not observe unrealized capital gains in mutual funds, we bind our estimates for taxable UCG under the alternative assumptions that mutual fund wealth is either 0% or 100% appreciated. Panel (b) depicts the distribution of unrealized capital gains in non-taxable real estate. Unrealized capital gains in taxable assets are concentrated at the very top of the wealth distribution. In contrast, unrealized capital gains in non-taxable real estate (the largest category of which is primary residences) are much more evenly distributed throughout the wealth distribution. Appendix Figure A3 compares these distributions to the distributions of realized and inherited capital gains. Panel (c) depicts the ratio of unrealized capital gains in taxable assets to total household net worth, which increases sharply at the top of the wealth distribution. Panel (d) shows a corresponding figure for the ratio of unrealized capital gains in non-taxable real estate to net wealth. Since household wealth at lower percentiles of net wealth is disproportionately in the form of (non-taxable) primary residences, this ratio is declining across the wealth distribution.

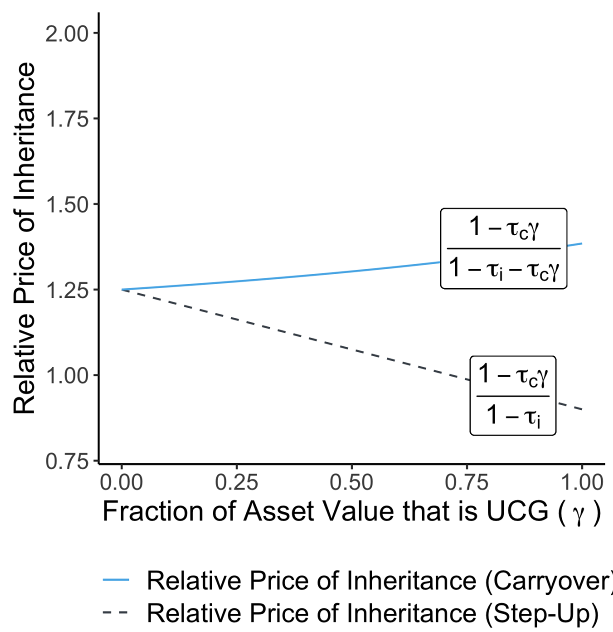
Figure 2: Inherited Capital Gains Across the Wealth Distribution, 2004



Note: This figure displays the cumulative distribution of inherited capital gains during 2004 with respect to the giver's household's percentile of net worth in 2004. Panel (a) displays this distribution for taxable assets — listed and unlisted stock and taxable real estate. Note that we are unable to observed inherited capital gains for mutual funds. Panel (b) shows a corresponding figure for the distribution of non-taxable real estate. The distribution of inherited capital gains in taxable assets is very concentrated at the top of the wealth distribution, even more so than the distribution of unrealized or realized capital gains (see Figure 1 and Appendix Figure A3). In contrast, inherited capital gains in non-taxable real estate (the largest category of which is primary residences) are much more evenly distributed throughout the wealth distribution.

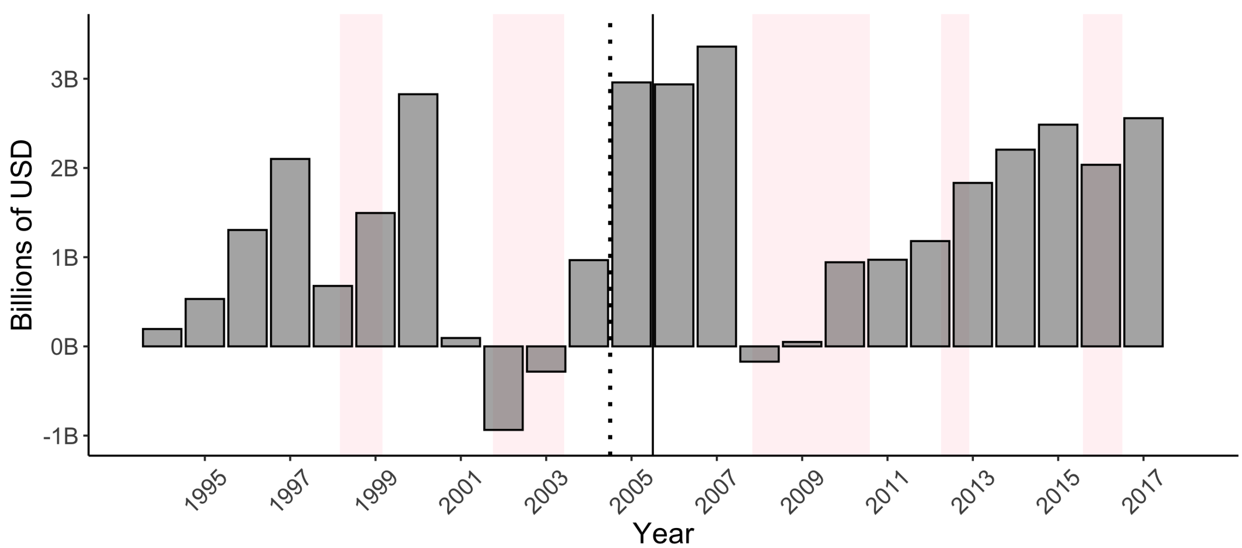
Figure 3: How the Removal of Step-Up Affects the Relative Price of Inheritance





Note: This figure demonstrates how the change from a policy of stepped-up basis to a policy of carryover basis increases the cost of inheritance (defined in Section 4.2 as $c_I = \tau_I + \theta\tau_c\gamma$), and thus how it affects the price of inheritance relative to present consumption ($\frac{1-c_R}{1-c_I}$). Panel (a) depicts how a change from step-up to carryover basis changes the relationship between the fraction of asset value that is unrealized capital gains (γ) and the cost of inheritance c_I: the flat line equal to the statutory inheritance tax rate (the dashed line τ_I) rotates to now depend steeply on γ (the solid light blue line $\tau_I + \tau_c\gamma$). Panel (b) depicts how this change in the cost of inheritance affects the relative price of inheritance in terms of consumption: $(1 - c_R)/(1 - c_I)$. The magnitude of the relative price change depends on the fraction of asset value that is unrealized capital gains: portfolios with no unrealized capital gains ($\gamma = 0$) experience no change in relative prices, while portfolios with large amounts of γ experience the largest price change. The figure is parameterized with the values $\tau_i = 20\%$ and $\tau_c = 28\%$, which were the top tax rates in place in Norway from 1994 to 2008.

Figure 4: Aggregate Net Capital Gains Realizations, 1994-2018



Note: This figure depicts the total amount of net capital gains (i.e., capital gains minus capital loss) realized in Norway from 1994 to 2018, in terms of 2018 USD. The light pink shading denotes recessionary periods, as classified by the OECD. The dotted black line before 2005 indicates the announcement of step-up's impending removal for stock. The solid black line before 2006 indicates the implementation of step-up's removal for stock. Net capital gains realizations tripled between 2004 and 2005: from \$0.97 billion in 2004 (6.24 billion nominal NOK) to \$2.96 billion in 2005 (19.4 billion nominal NOK).

Table 2: Summary Statistics of Key Variables for Baseline Difference-in-Difference Analysis

	Control Group	Treatment Group
Panel A. Per-Dollar Cost of Inheritance for Stock (c_I)		
During Step-Up (1999-2005)	0.20	0.20
	(0.00)	(0.00)
During Carryover (2006-2008)	0.20	0.33
	(0.00)	(0.07)
Percent Change (Post vs. Pre)	0.00	0.65
Panel B. Yearly Realized Capital Gains		
Before Announcement (2000-2004)	3,623.05	2,438.98
	(122,075.60)	(104,115.20)
After Announcement (2005-2008)	5,132.62	5,479.09
	(80,933.24)	(106,790.30)
Percent Change (Post vs. Pre)	0.42	1.25
Panel C. Covariates		
Age in 2005	50.11	52.18
	(15.81)	(16.69)
Indicator for any unlisted holdings	0.60	0.55
	(0.49)	(0.50)
Number of Individuals	134,242	344,396

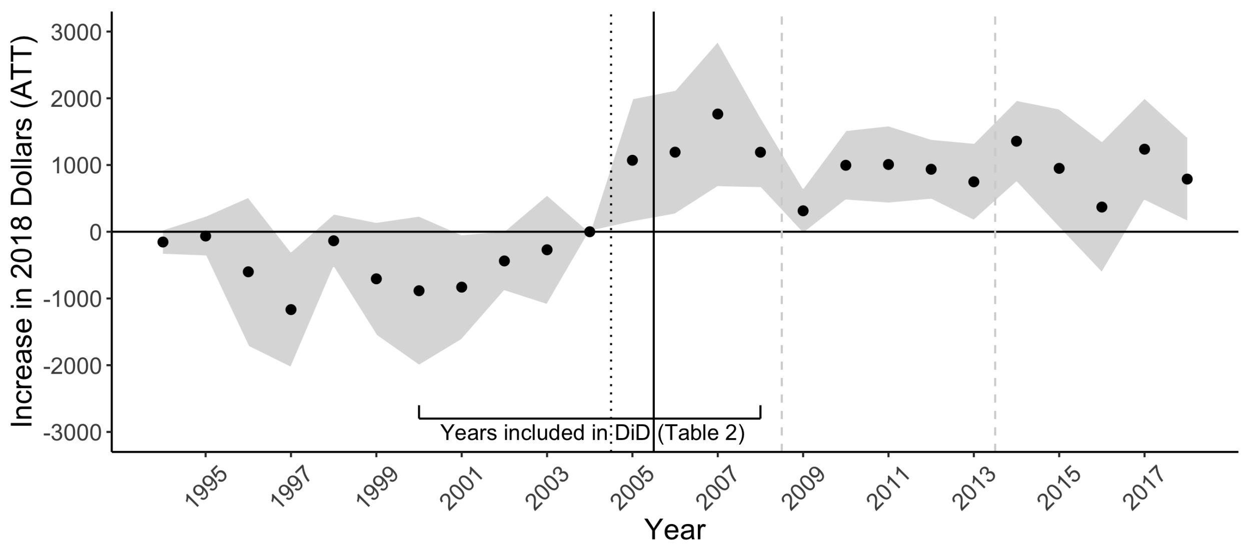
Note: This table depicts means and standard deviations (in parentheses) for our measure of the "per-dollar cost of inheritance" for stock (Panel A), our outcome variable of yearly realized capital gains from any asset class (Panel B), and several covariates (Panel C) for the treatment and control groups in our baseline analysis of the removal of step-up for stock. The per-dollar cost of inheritance is defined as $c_I = \tau_I + \theta_t \tau_c \gamma$ in Section 4.2. Before 2006, step-up is in place ($\theta = 0$), so the total cost is simply the statutory inheritance tax rate (τ_I). Once step-up is removed ($\theta = 1$), costs begin to depend on the amount of appreciation in each individual's stock portfolio. We end our baseline analysis in 2008 because inheritance tax rates change in 2009, but Appendix Figure A6 depicts c_I over a longer time horizon. Realized capital gains are the individual's net realized capital gains, as reported on his or her yearly income tax return, which includes realized capital gains from stock and from all other taxable asset categories. Individuals are included in this table is they have some stock holdings at the beginning of 2005, and are sorted into the treatment group if their stock portfolio has net positive unrealized capital gains ($\Gamma_i \geq 0$) at the beginning of 2005. Values are in 2018 USD.

Table 3: Realization Responses to the Removal of Step-Up for Stock, 2000-2008

Outcome: Yearly Realized Capital Gains	(1)	(2)	(3)
TREAT × POST (β_1)	0.461***	0.483***	0.492***
	(0.109)	(0.094)	(0.073)
Covariates:			
Indicator for any unlisted holdings		X	X
Decade Age Bin			X
Number of Observations	4,124,605	4,124,605	4,011,557
Implied Treatment Effects:			
$\exp(\beta_1) - 1$	.59	.62	.64
Average Increase in Realizations (ATT)	\$2,023	\$2,163	\$2,152
Total Increased Realizations (ATT $\times N_{TREAT}$)	\$697.02 mil	\$745.23 mil	\$741.43 mil

Note: This table shows our baseline difference-in-difference estimates, which estimate Equations 1 and 3 using Poisson Pseudo Maximum Likelihood. Observations are at the individual-year level. Individuals are included in these regressions if they owned stock at the beginning of 2005. The outcome variable is net realized capital gains from any asset class, as reported yearly on each individual's tax return. The indicator "TREAT" equals 1 if the individual is exposed to the removal of step-up (i.e., has positive appreciation in his or her stock portfolio at the beginning of 2005). The indicator "POST" equals 1 for years after 2004, and 0 before then. The first row of the table presents the coefficient from the Poisson regression. Standard errors clustered at the individual level are presented in parentheses below. To interpret our coefficients in terms of percent changes in the outcome variable, we need to exponentiate the coefficient and subtract by one. We form an estimate of the ATT from regression parameters as described in Equation 2. We can calculate the total increase in taxable realizations (relative to counterfactual) by multiplying the ATT by the number of individuals in the treatment group (N_{TREAT}). Values are in 2018 USD. Column (1) includes no covariates. Column (2) includes an indicator for whether a portfolio includes any unlisted stock (including holding companies). Column (3) includes additional indicators for an individual's 10-year age bin. Covariates are included in the regressions as fully saturated indicators. We end our baseline analysis in 2008, before the 2009 decrease in inheritance tax rates. Figure 5 displays year-specific event study estimates over a longer time horizon.

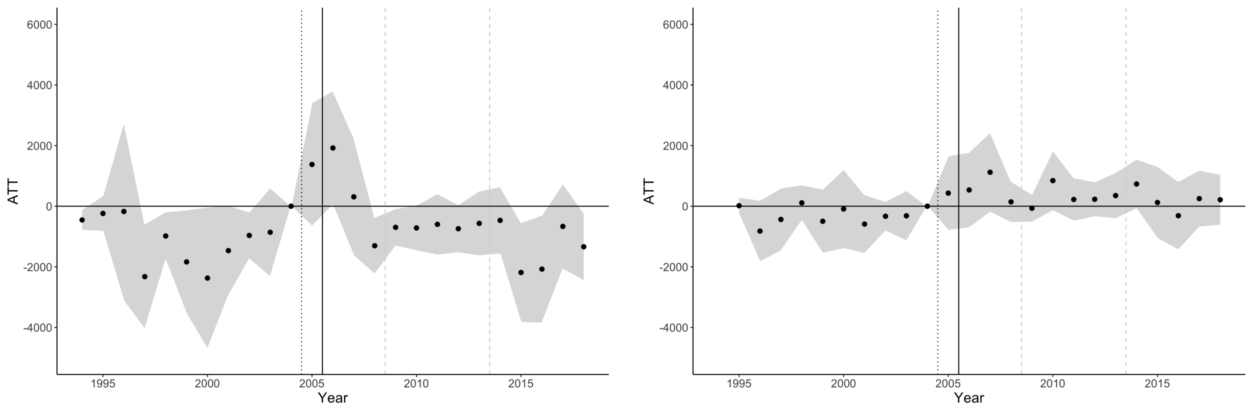
Figure 5: Average Yearly Increase in Realized Capital Gains (Event Study)



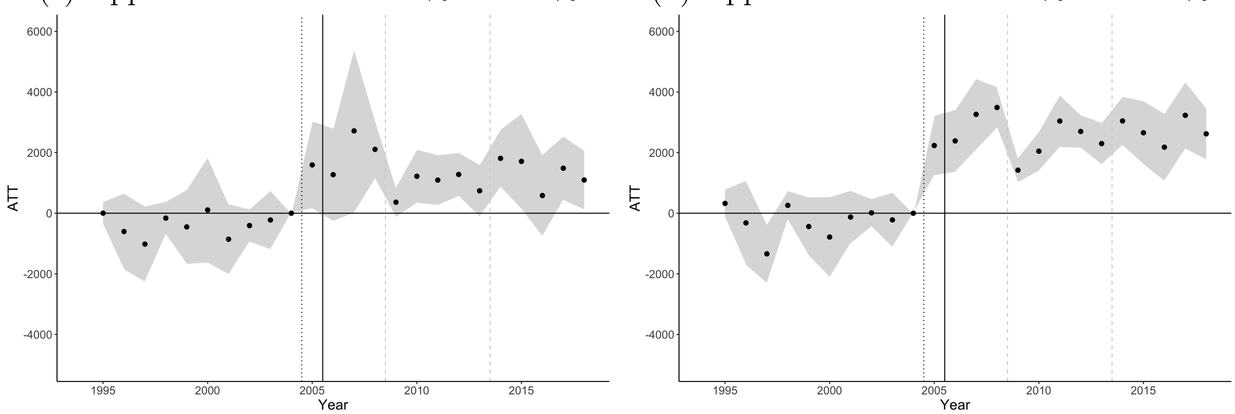
Note: This figure shows the average increase in realized capital gains among all treated individuals, in response to the removal of stepped-up basis for stock. Treated individuals are defined as those with any positive amount of unrealized capital gains in their stock portfolio at the beginning of 2005. The control group is individuals with zero capital gains, or net capital losses, in their stock portfolio at the beginning of 2005. The y-axis presents the level of the estimated treatment effect in 2018 USD. The dotted black line before 2005 indicates the announcement of step-up's impending removal for stock. The solid black line before 2006 indicates the implementation of step-up's removal for stock. In contrast to other figures in the main text of this document (and to the baseline difference-in-difference estimates presented in Table 3), this figure extends beyond 2008, and thus includes multiple reforms to the inheritance tax rate. The dashed gray line before 2009 indicates a fall (by half) of the statutory inheritance tax rate. The dashed gray line before 2014 indicates the full removal of the inheritance tax, and the beginning of a period of increasing capital gains tax rates. See Appendix B for more detail on these later policy changes. Our baseline difference-in-difference estimates (Table 3) only include data from 2000 to 2008, in order to end the analysis before the 2009 inheritance tax rate change.

Figure 6: Average Yearly Treatment Effect (Event Study), by Bin of Treatment Intensity

(a) Appreciation between 0 and 25% (b) Appreciation between 25% and 50%



(c) Appreciation between 50% and 75% (d) Appreciation between 75% and 100%



Note: This figure shows heterogeneity in the average responses to the removal of stepped-up basis for inherited stock. The y-axes present the level of the estimated treatment effect in 2018 USD. Panel (a) depicts the response for individuals whose stock holdings, at the beginning of 2005, have a ratio of unrealized capital gains to value (γ) between zero and 0.25. Panel (b) shows the corresponding figure for individuals with 2005 stock holdings with γ between 0.25 and 0.5. Panel (c) is the corresponding figure for those with γ between 0.5 and 0.75. Panel (d) is the corresponding figure for those with γ between 0.75 and 1. In all panels, the control group is individuals whose stock portfolios in 2005 had zero capital gains or net capital losses. The dotted black line before 2005 indicates the announcement of step-up's impending removal for stock. The solid black line before 2006 indicates the implementation of step-up's removal for stock. The dashed gray line before 2009 indicates a fall (by half) of the statutory inheritance tax rate. The dashed gray line before 2014 indicates the full removal of the inheritance tax, and the beginning of a period of increasing capital gains tax rates.

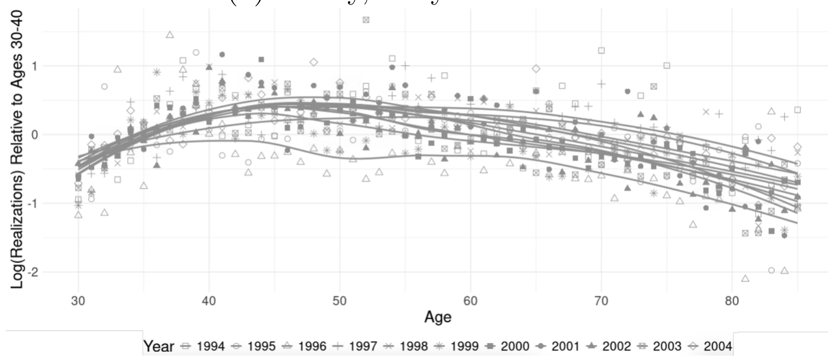
Table 4: Realization Responses to the Removal of Step-Up by Parenting Status, 2000-2008

Outcome: Yearly Realized Capital Gains *Sample*:	(1) All	(2) Parents	(3) Non-Parents
TREAT × POST (β_1)	0.461***	0.506***	0.183
	(0.11)	(0.06)	(0.30)
Number of Observations	4,004,889	3,446,931	557,958
Implied Treatment Effects:			
$\exp(\beta_1) - 1$	.59	.66	.20
ATT	$2,023	$2,262	$721
Increased Tax Revenue (ATT $\times N_{TREAT}$)	$697.02 mil	$642.47 mil	$33.08 mil

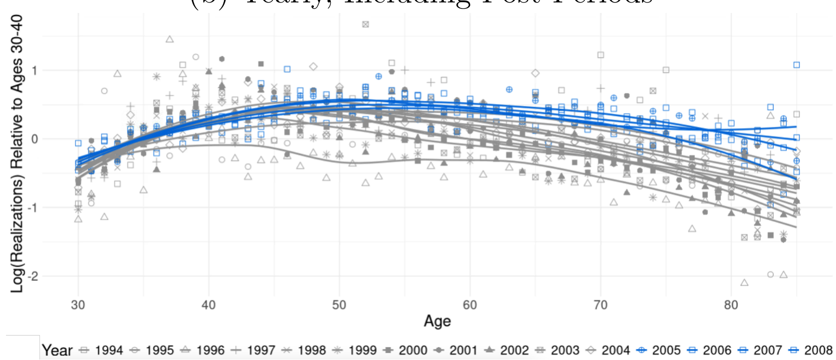
Note: This table presents the difference-in-difference estimates as in Table 3, but here breaks out the sample by whether an individual has children. Column (1) includes the entire sample of individuals who owned stock at the beginning of 2005 (a reproduction of the first column of Table 3). Columns (2) and (3) break the sample into those individuals who are and are not parents, respectively. Parenting status is defined by the presence of any children in the Norwegian register data from 1994-2018. Individuals are therefore flagged as parents prior to the birth (and potentially after the death) of their children, as long as those children are observed between 1994 and 2018. Estimation uses Poisson Pseudo Maximum Likelihood. Observations are at the individual-year level. The outcome variable is net realized capital gains from any asset class, as reported yearly on each individual's tax return. The indicator "TREAT" equals 1 if the individual is exposed to the removal of step-up (i.e., has positive appreciation in his or her stock portfolio at the beginning of 2005). The indicator "POST" equals 1 for years after 2004, and 0 before then. The first row of the table presents the coefficient from the Poisson regression. Standard errors clustered at the individual level are presented in parentheses below. To interpret our coefficients in terms of percent changes in the outcome variable, we need to exponentiate the coefficient and subtract by one. We form an estimate of the ATT from regression parameters as described in Equation 2. We can calculate the total increase in taxable realizations (relative to counterfactual) by multiplying the ATT by the number of individuals in the treatment group (N_{TREAT}). Values are in 2018 USD. We end our analysis in 2008, before the 2009 decrease in inheritance tax rates.

Figure 7: Relationship between Age and Realizations, by Year

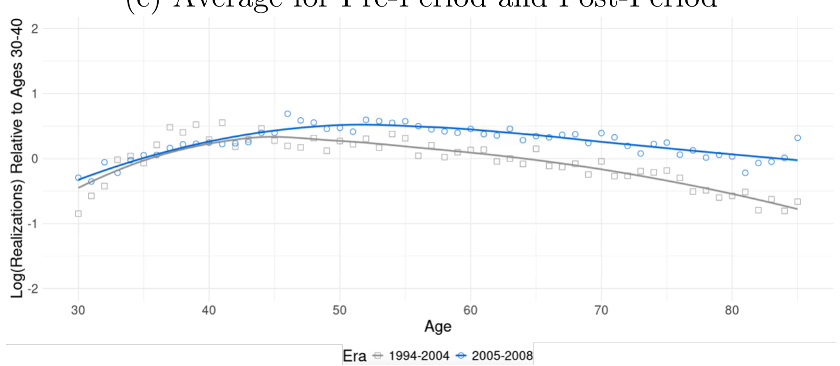
(a) Yearly, Only Pre-Periods



(b) Yearly, Including Post-Periods



(c) Average for Pre-Period and Post-Period



Note: This figure plots the log of average realizations by age and year from 1994-2008. Y-axis values are normalized to the average value for the 30-to-40 year-olds in the same year. The points are year- (or era-) and age-specific logged averages. The smoothed lines depict local nonlinear regressions (LOESS) of logged average realizations on age for each year or era. Panel (a) includes only years prior to 2005. Panel (b) adds in the years from 2005 to 2008 (in blue). Panel (c) includes the same years but presents averages for 1994-2004 and 2005-2008, rather than by year. Panels (b) and (c) reveal that older individuals realized relatively more since 2005. We end our analysis in 2008, before the 2009 decrease in inheritance tax rates.

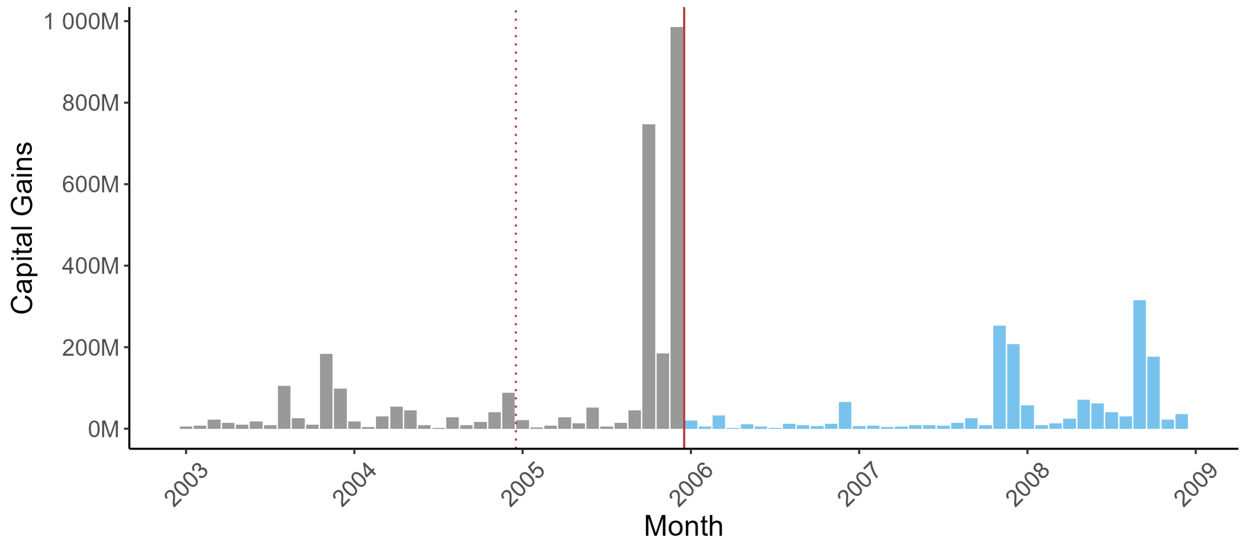
Table 5: Realization Responses by Age, 2000-2008

Outcome: Yearly Realized Capital Gains	(1)	(2)	(3)	(4)
Ages in "OLD" Treatment Group:	50-59	60-69	70-79	80-89
OLD × POST (β_1)	0.229**	0.342***	0.503***	0.638***
	(0.086)	(0.072)	(0.084)	(0.133)
Number of Observations	2,467,859	2,182,601	1,881,847	1,693,930
Implied Treatment Effects				
$\exp(\beta_1) - 1$	0.26	0.41	0.65	0.89
ATT	\$1,365	\$1,704	\$1,945	\$1,681

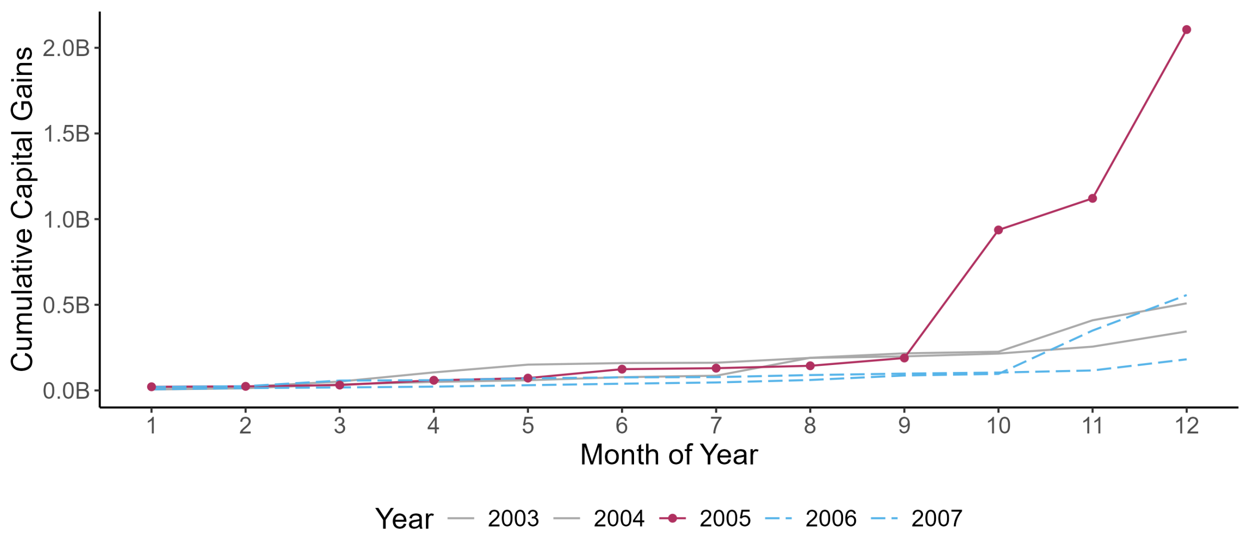
Note: This table presents the difference-in-difference estimates for our empirical strategy which compares older age groups to a control group of individuals aged 30-49. Specifically, we estimate Equation 4 using Poisson Pseudo Maximum Likelihood. Observations are at the individual-year level. Individuals are included in these regressions if they owned stock at the beginning of 2005. The outcome variable is net realized capital gains from any asset class, as reported yearly on each individual's tax return. The indicator "OLD" equals 1 if the individual is in the age group below the numerical column header, and equals 0 if the individual is in the 30-49 age group. Other age groups are not included. The indicator "POST" equals 1 for years after 2004, and 0 before then. The first row of the table presents the coefficient from the Poisson regression. Standard errors clustered at the individual level are presented in parentheses below. To interpret our coefficients in terms of percent changes in the outcome variable, we need to exponentiate the coefficient and subtract by one. We form an estimate of the ATT from regression parameters as described in Equation 2. We can calculate the total increase in taxable realizations (relative to counterfactual) by multiplying the ATT by the number of individuals in the treatment group (N_{TREAT}). Values are in 2018 USD. We end our analysis in 2008, before the 2009 decrease in inheritance tax rates.

Figure 8: Anticipatory Inheritance Transfers in Late 2005

(a) Capital Gains in Inherited Stock, by Month

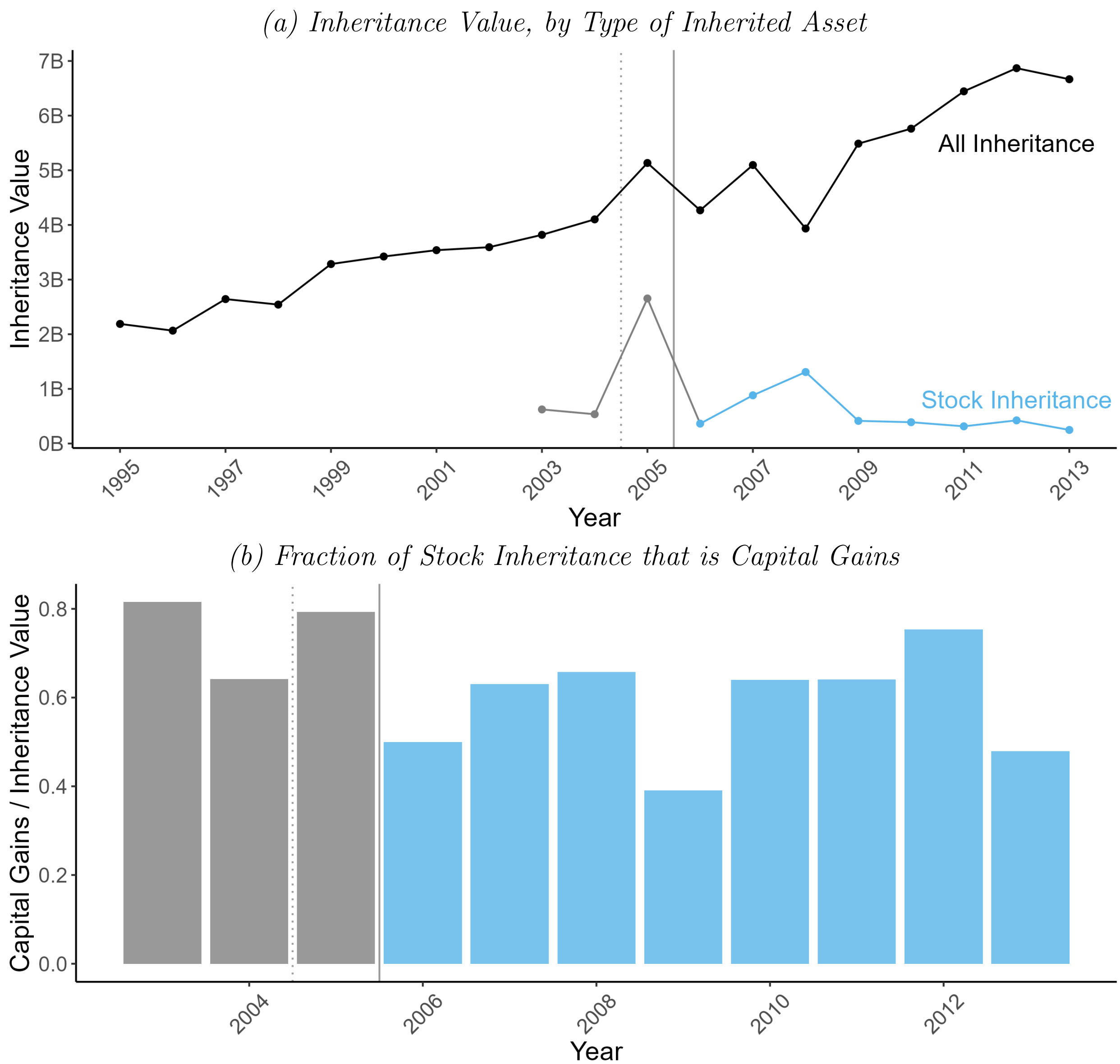


(b) Cumulative Capital Gains in Inherited Stock, by Month of Year



Note: This figure demonstrates the anticipatory transfers that occurred in late 2005, which were triggered by the announcement that step-up would be removed for stock on January 1, 2006. Transfers made prior to January 2006 would still be subject to stepped-up basis treatment. Panel (a) shows that the amounts of capital gains transferred in inheritance in the later months of 2005 were large outlier values compared to monthly values before or since. The dotted red line before 2005 indicates the announcement of step-up's impending removal for stock. The solid red line before 2006 indicates the implementation of step-up's removal for stock. Panel (b) displays the cumulative value transferred over each year — the amount of capital gains transferred in inheritance in 2005 was much more than the prior or immediately subsequent years. Values are in 2018 USD.

Figure 9: Aggregate Inheritance Trends Following the Removal of Step-Up for Stock



Note: Panel (a) shows that the value of all inheritance (i.e., of cash or of any other asset) is growing over time, from \$2.19 billion in 1995 to \$6.67 billion in 2013, the last year prior to the abolition of the inheritance tax. The value of stock transferred in inheritance, on the other hand, does not show consistent growth after step-up is removed in 2006. Values are in 2018 USD. Furthermore, Panel (b) shows that the fraction of inherited stock value that is capital gains falls somewhat following the removal of step-up for stock. Combined, these two panels suggest that although the total value of inheritance is growing in Norway over time, an increasingly small fraction of that inherited value is composed of capital gains in stock. The dotted gray line before 2005 indicates the announcement of step-up's impending removal for stock. The solid gray line before 2006 indicates the implementation of step-up's removal for stock.

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Appendix

A Background on Capital Gains Taxation

A.1 What are Capital Gains?

Capital gains refer to the appreciation of an asset (such as stocks, real estate, or other investments) during the asset owner's holding period. When the current price of an asset exceeds the owner's original purchase price (which is called the owner's "basis" in the asset)[46], that owner has capital gains:

$$\text{Capital Gains} = \text{Current Price - Basis}$$

The above formula holds even if the asset has fallen in value since the owner's initial purchase, in which case the owner is said to have a capital loss in the asset. For sake of brevity we will use the term "capital gains" to include net capital gains and losses throughout this paper. Note that our definition of capital gains includes all gains accumulated during an investor's holding period, whether or not that gain is recognized for tax purposes upon sale or another realization event. Throughout this paper, we will therefore need to distinguish between "realized capital gains" and "unrealized capital gains," depending on whether the underlying asset experiences a taxable realization event. We will also refer to those not-yet-realized capital gains that pass in gift or inheritance transactions as "inherited capital gains."

Capital gains are a form of income tied to asset ownership. This is not to say that all capital gains are "capital income," as is usually distinguished from labor income in economic theory. Many capital gains, such as those built up in a business by an entrepreneur, or by renovating real estate, are clearly a return to the owner's human capital and labor effort. Mapping optimal tax theory of capital and labor income onto the legal category of capital gains income is outside the scope of this paper.[47] We take the existence of capital gains as a

[46]The owner's basis in the asset would also include any costs associated with his acquisition of the asset, such as commissions or fees.

[47]Several seminal results in the theoretical tax literature contend that the optimal tax rate on capital income is zero (Chamley, 1986; Judd, 1985; Atkeson et al., 1999), although this view has also been widely challenged (see, e.g., Aiyagari (1995); Straub and Werning (2020)).

distinct tax base with a positive tax rate as given and consider changes to that capital gains tax base.

A.2 Taxation of Capital Gains

A.2.1 A Realization-based Tax

Although capital gains accrue continuously over time, they are typically only taxed when *realized* through the sale of the underlying asset.[48] Therefore, an individual may own a stock for many years and accrue a large amount of *unrealized* capital gains in the stock. But until he sells the stock (or until another realization event specified by the tax authority occurs), he will not owe tax on that capital gain.

The realization requirement is widely believed to be due to the administrative issues with, and political opposition to, taxing capital gains as they accrue.[49] Taxing capital gains at the time of sale sidesteps the need for the government to value unsold assets and aligns the timing of the tax with when the investor has liquid wealth to pay the tax bill. Despite the near-consensus[50] in the academic literature in favor of accrual-based taxation (at least in theory), policymakers and the public seem to favor a realization requirement, as evidenced both by the lack of global policy experimentation (no country has implemented a broad accrual system), and recent survey evidence in the US (Liscow and Fox, 2022).

A.2.2 Possible Tax Treatments of Never-Realized Capital Gains

As long as the capital gains tax remains a realization-based tax, a choice will need to be made as to how to treat capital gains that are never realized by the original owner, and instead pass to that owner's heirs as gifts or inheritances. There are three broad alternatives:

[48] Although we are not aware of any country with a broad accrual-based system for individual capital gains taxation, accrual-based taxation is sometimes available in limited form. For example, Section 475 of the US Tax Code requires mark-to-market accounting for securities dealers and offers an election into mark-to-market treatment for commodities dealers and traders.

[49] Descriptions of the realization rule as a "concession [...] to the exigencies of a given situation" date at least as far back as Haig (1921).

[50] Aguiar et al. (2024) is a notable recent exception to this consensus. They argue that when some capital gains accrue due to changes in the discount rate, an optimal tax base would have elements of both an accrual-based and realization-based system.

1. Stepped-up Basis

Under a system of stepped-up basis, inherited capital gains disappear from the capital gains tax base. No capital gains taxes are due at the time of the inheritance transfer, and if the heir subsequently realizes the asset, she pays capital gains tax only on appreciation that took place during her own holding period. Formally, this result is accomplished by deeming the heir's basis in the asset to be "stepped up" to the market value at the time of the inheritance transfer.

2. Carryover Basis

Under a system of carryover basis, the heir inherits the original owner's unrealized capital gains when she inherits the associated asset. No capital gains taxes are due at the time of the inheritance transfer, but if the heir subsequently realizes the asset, she pays capital gains tax on all appreciation since the acquisition by the original owner. Formally, this result is accomplished by setting the heir's basis in the asset equal to her predecessor's basis.

3. Forced Realization

If the inheritance transfer is deemed to be a realization event, capital gains that the original owner has not yet realized will be realized and taxed at the time of inheritance. If the heir subsequently realizes the asset, she will then pay capital gains tax only on appreciation that took place during her own holding period. Formally, this result is accomplished by acting as if the asset was sold by the original owner to the heir for market value at the time of transfer. The heir's basis in the asset going forward is therefore the market value at the time of the transfer.

Each of these alternatives is currently in use in different countries around the globe. Canada, for example, imposes forced realization events to tax as-yet-unrealized capital gains that are transferred as gifts or inheritance. The UK also imposes forced realization, but only for *inter vivos* gifts. In Australia and Mexico, inheritors of capital assets carry over their predecessor's unrealized capital gains, which are taxed when the asset is next realized (Joint Committee on Taxation, 2008). Carryover basis is also in place in the US for *inter vivos* transfers. Deathtime transfers, on the other hand, are subject to stepped-up basis in the US, the UK, and 10 other OECD countries (OECD, 2021).

A.2.3 Interaction with an Inheritance or Estate Tax

Although repealing stepped-up basis and inheritance taxes are frequently considered in tandem, an inheritance tax is not a perfect substitute for exempting inherited capital gains from the capital gains tax base. Inheritance transfer taxes apply to the market value of the bequest, regardless of any unrealized capital gains. Under stepped-up basis, an investor who sells a highly appreciated asset the day before death and leaves the resulting cash as inheritance is subject to both the capital gains tax and the inheritance tax, whereas a dying investor who bequests the unrealized asset itself pays only the inheritance tax (if any). In Section 4, we clarify formally how the inheritance tax rate and the tax treatment of inherited capital gains each affects the cost of inheritance.

B Background on Norway

B.1 Taxation in Norway, 1992-2018

Norwegians pay taxes on their worldwide labor income, capital income, and wealth. Labor income is subject to progressive taxation with a top marginal rate of 53.2–54.3% (including social security contributions) during our period of study. Net wealth in excess of an exemption amount faces a yearly tax of 0.85–1.30% during our period.[51]

B.1.1 Capital Gains and Inheritance Taxation

Norway has a flat capital gains tax on realized capital gains. From 1992 to 2013, all (net) realized capital gains were taxed at a rate of 28%.[52] Since 2014, there have been several rate

[51]The wealth tax exemption has ranged from $19,555 to $162,800 (2018 USD) during our period of study. In general, exemptions have grown more generous and wealth tax rates have fallen over our period of study. See Thoresen et al. (2022) for a detailed description of Norway's wealth tax. Interestingly, Norway collects a smaller share of total tax revenue from property and wealth taxation than is typical among OECD countries (OECD, 2018).

[52]This rate is also applied to other forms of capital income, such as interest income. Capital losses can be used to fully offset capital gains. Net losses can be used to offset ordinary income: $1 of capital losses offsets only $0.72 of ordinary income, and cannot be applied against social security contributions or top marginal tax rates.

changes, including the introduction of separate rates for the realization of capital gains from stock and non-stock assets. These rate changes are depicted in Figure A4. Throughout our study period, the capital gains tax has had a flat rate structure, so that capital gains tax rates do not depend on individual income.

Prior to its abolition in 2014, Norway's integrated gift and inheritance transfer tax taxed the receipt of gifted or inherited assets through a rate structure that depended on the cumulative amount of the gifts/inheritance received, and the relationship between the giver and the recipient. Other than a small yearly gift exemption, no distinction was made between *inter vivos* gifts and bequests at the time of death. Appendix Table A1 provides details on the rates and brackets over time. Figure A4 depicts the top rate for gifts and inheritances between close relatives: a rate that fell from 20% to 10% in 2009.[53] In 2014, the gift and inheritance tax was abolished, so all rates for subsequent years are set to 0%.

B.1.2 Stepped-Up Basis Policy

During our period of study, Norway moved from a system of stepped-up basis for inherited capital gains to a system of carryover basis. This change was accomplished in two steps. In 2006, carryover basis was introduced for appreciated gifted or inherited stock (including equity-based mutual funds). Recipients of gifted or inherited stock from 2006-2013 continued to be subject to inheritance tax on value of the stock they received, but now were also subject, if and when they sold the stock, to capital gains tax on the amount of appreciation during the original owner's holding period. Losses, however, were not permitted to be carried over in the 2006 reform – so stocks with losses continued to be subject to a "step-down" in basis. In 2014, concurrent with the abolition of the gift and inheritance tax, carryover basis was introduced for other assets, including stocks with losses. Therefore, since 2014, recipients of gifted and inherited assets do not pay inheritance tax at the time of the receipt, but are subject to capital gains tax on appreciation from the original owner's holding period when they sell the inherited asset.

[53] The definition of close relatives includes parents and children of the giver. Transfers between spouses were not taxed. Transfers between individuals who are not close relatives were subject to a higher top rate in every year: 30% before 2009, and 15% until 2014.

Certain types of real estate are exempt from capital gains taxation in Norway if the owner's use of the property and length of ownership meets certain requirements. Most prominently, residential property is not subject to capital gains tax if the seller has owned and lived in the property as his primary residence for 12 of the last 24 months. Vacation homes can be similarly exempt if the seller has used it as his own vacation home for five of the last eight years. Buildings on a farm may be exempt if the farm has been lived on for over ten years. In these cases in which the original owner could have sold the property without capital gains tax, the heir's basis continues to be stepped up to the assumed sales value at the time of acquisition, even after 2014.

B.1.3 Legislative History & Anticipation of Step-Up's Removal

Norway's policy of stepped-up basis for gifted and inherited assets was first established by Norwegian Supreme Court rulings in the 1920s. Various government committees sporadically proposed moving to a system of carryover basis (proposals were made in 1975, 1981 (for gifts only), 1989 and 2000), but no changes were enacted until 2006 (Zimmer, 2014).

In 2005, it was announced that step-up would be removed for stock (including unlisted stock and equity-based mutual funds). The 2006 removal of stepped-up basis for inherited stock was passed as an afterthought to a wider reform of capital taxation. The 2006 tax reform changed the exemption rules for dividend and capital gains taxes in a way that made these exemptions more dependent on owners' cost basis (see Section B.1.4). Because of concern that owners might transfer shares as gifts to their relatives to get the increased dividend exemptions that would accompany stepped-up basis, subsequent legislation was passed that abolished stepped-up basis for shares starting in 2006. Although the main components of the 2006 tax reform were formally announced in March 2004 and approved in June of that year, the removal of step-up was not part of the June legislation and was not definitively decided until later in the year (Norwegian Ministry of Finance, 2004). In a search through the Norwegian media library (Nasjonalbiblioteket) for the terms "kontinuitetsprinsippet" and "diskontinuitet" (which are the Norwegian terms corresponding to carryover and stepped-up basis, respectively), we do not find any press coverage regarding the impending removal of step-up until early 2005.

The 2014 removal of step-up for other assets was closely tied to the abolition of the inheritance tax. Right-wing parties campaigned on abolishing the inheritance tax during the 2013 election, which they won in September 2013. Therefore from September 2013, it was reasonably certain that the inheritance tax would be repealed, although the exact timing of the repeal initially remained uncertain. The rules accompanying that repeal, which included the removal of step-up for non-stock assets, were announced as part of the government's yearly budget in October 2013.

B.1.4 Other Aspects of the 2006 Reform

As mentioned above, the removal of step-up for stock came about due to a larger overhaul of exemptions for capital gains and dividend taxation. The overall reform was primarily motivated by a need to prevent private business owners from relabeling labor income as dividends, which had become endemic to the old system. Under the old system (called "RISK"), both capital gains and dividends were given generous exemptions that corresponded to the amount of retained earnings held in the underlying firm. The goal of this exemption was to avoid double-taxation on corporate earnings which would have already been subject to the corporate tax. Furthermore, dividend distributions were subject to a yearly limit that corresponded to the amount of retained earnings in the firm at the beginning of the year. Because the exemption amount and dividend cap were thus linked, the effect was that almost no dividends were taxed prior to 2006. From 2006, the same dividend cap remained in place, but the exemption calculations for dividends and capital gains were made less generous going forward. Henceforth, investors would accumulate exemptions based on how their costs basis would grow with the normal rate of return.

Previous work has documented that the 2006 reform led to a large fall in dividend distributions (Alstadsæter and Fjæli, 2009; Alstadsæter et al., 2014, 2019b). However, we do not think that this aspect of the reform is likely to be linked to the increase in capital gains realizations we document later in this paper (Section 5). First of all, our results in Section 5 show a response starting in the announcement year of 2005, before the exemption calculations changed.[54] Secondly, the old exemption amounts remained usable for realizations after 2006 – only the accumulation of future exemption amounts changed.[55] Therefore, there was no immediate change to the tax calculation of realizing capital gains in 2005 vs. early 2006 – the same exemption amounts would be available both times. Finally, we do not think there is any incentive for investors to shift from dividend distributions to capital gains realizations. Both before and after the reform, dividends and capital gains were subject to the same tax rate and granted the same exemption amounts. Investors should therefore be "neutral" between the two methods of extracting money from the firm both before and after the reform (Sørensen, 2005).[56] We therefore do not expect any of our documented increase in capital gains realizations (Section 5) to be a shift from dividends to capital gains realizations.

C Model Proofs

Proof of Proposition 1 (Step-Up Depresses Realizations) *If $\mu > 0$ and the substitution effect dominates the income effect, replacing stepped-up basis with carryover basis (i.e., a change from $\theta = 0$ to $\theta = 1$) will weakly increase the amount C realized for present consumption.*

Set-Up: Call p the relative "take-home rate" of bequests: $p \equiv \frac{1-c_R}{1-\mu c_I} = \frac{1-\tau_c\gamma}{1-\mu(\tau_I+\theta\tau_c\gamma)}$ so that the budget constraint can be rewritten: $C + pB = W(1-\tau_c\gamma)$. In this reformulation, it is clear that the effect of the removal of step-up (increasing θ from 0 to 1) when $\mu > 0$ is to weakly increase p, the relative price of bequests. This increase is "weak" because a change in θ has no effect if $\gamma = 0$. We want to show that the uncompensated cross-price elasticity between C and B is weakly positive, as long as the income effect is not too large relative to the substitution effect.

Proof. The law of demand gives us that the Hicksian own-price elasticity of demand is negative ($e^H_{cc} \leq 0$). Since Hicksian demand is homogeneous of degree zero in prices, the sum

[54] It is important to recall that our realization results look at the outcome of *taxable* realized capital gains, and so do not include any (tax-free) company transformations occurring under Transition Rule E.

[55] See The Norwegian Tax Administration (n.d.b): "For shares acquired before 1 January 2006, the input value is adjusted for accumulated RISK."

[56] See Bjerksund and Schjelderup (2021a,b) for discussions of how exemptions and loss limitations affect the capital gains realization decision in Norway.

of all Hicksian cross price elasticities is zero. Therefore, in a 2 good model, the sole Hicksian cross-price elasticity is positive. By the Slutsky equation, the total (Marshallian) elasticity of demand equals the Hicksian elasticity of demand minus the budget share times the income elasticity of demand: $e^M_{cb} = e^H_{cb} - s_b\eta_c$. We know $s_b \geq 0$ and, if realizations are a normal good, then $\eta_c > 0$. Therefore e^M_{cb} is positive as long as $s_b\eta_c < e^H_{cb}$. □

D Data Validation Appendix

Our paper uses transactions data to build new measures of unrealized and inherited capital gain at the individual-asset level. We present several validation checks on components of these novel measures below.

D.1 Validating End-of-Year Stock Holdings

From 2004, the Norwegian Shareholder Registry documents the ownership shares by Norwegian shareholders of AS and ASA (limited liability) companies. We can therefore compare the ownership amounts documented in the Shareholder Registry data with our calculation of individual shareholders' end-of-year holdings in these companies.

The two measures of ownership agree very closely: for AS and ASA companies subject to Shareholder Registry reporting, we are able to match 97% of Norwegian shareholders in our data into the Shareholder registry, and 99.6% of all matched person-year observations agree perfectly between the two datasources. That is, our transaction-based calculation of a given individual's number of shares owned at the end of the year in a given company usually agrees perfectly with the number of shares that the Shareholder Registry says that individual owns in that company. Residual disagreement (illustrated in Appendix Figure A1 Panel A) is probably due to issues with merging the two datasets, or limitations in our procedures to account for mergers and other company transformations.

D.2 Comparing Realized Capital Gains to Tax Returns

Our measures of unrealized and inherited capital gains are novel, and thus hard to compare to existing estimates. In contrast, taxable realized capital gain is a component of taxable income, and is documented each year in every individual's tax return. Validating our calculation for realized capital gain allows us to conclude that our measure of basis is sensible, and that the transactions data reflects realizations in the tax return.

However, we are not able to make an exact comparison between two measures of realized capital gain in stock. The individual tax returns report total taxable capital gains and losses and real estate-specific taxable gains and losses. Below, we compare net taxable realized capital gains from assets other than real estate to our measure of realized capital gains from directly-held stock (listed and unlisted), net of the rate-of-return deduction. These measures may differ if the individual realized capital gains in asset classes other than real estate or directly-held stock (e.g., from mutual funds). It may also differ (although probably only slightly) because individuals can deduct transaction costs (such as broker fees) from their taxable gain, and we do not observe these costs in our data.

Despite these known differences, the two measures of realized gain align well. 97% of all individuals in the tax data report net realized capital gain from non-real estate that is within rounding error (2000 NOK) from our calculation in the stock transactions data. Appendix Figure A1 Panel B illustrates the comparison.

D.3 Comparing Aggregate Values to Andresen (2022)

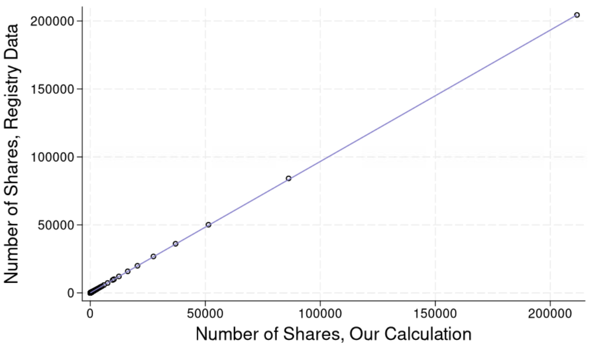
Andresen (2022) presents aggregate measures of unrealized capital gains for unlisted companies in Norway from 2006-2016. This Statistics Norway report is the only other work we are aware of that makes such a calculation. Andresen calculates unrealized capital gains as equal to the value of the unlisted stock minus basis and rate-of-return deduction. He predicts the market value of unlisted stock by multiplying book value times a book-to-price multiplier calculated from listed Norwegian companies. He has a direct measure of basis and rate-of-return deductions from the Norwegian Tax Authority (data we ourselves do not have access to), but acknowledges large problems and unlikely outliers in this measure.[57] Despite

[57] To quote from the report regarding the issue with the Norwegian Tax Authority's data on basis (not used in our own analysis): "The data on basis unfortunately has several weaknesses. In many cases, the basis values will be too large [...] Somewhat surprisingly, there is also a significant number of negative basis values in this data set, and for the years 2006-2009 this is so common that the sum of all basis values for

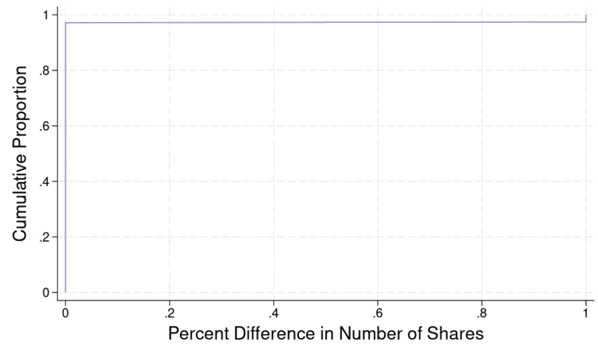
Figure A1: Validation Exercises

Panel A: End-of-Year Share Comparison

(i) Binned Scatter Comparison

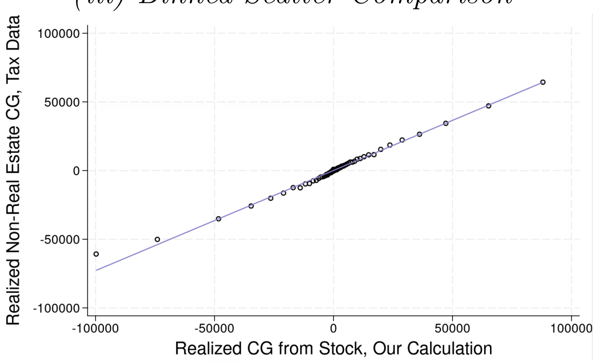


(ii) CDF of Disagreement

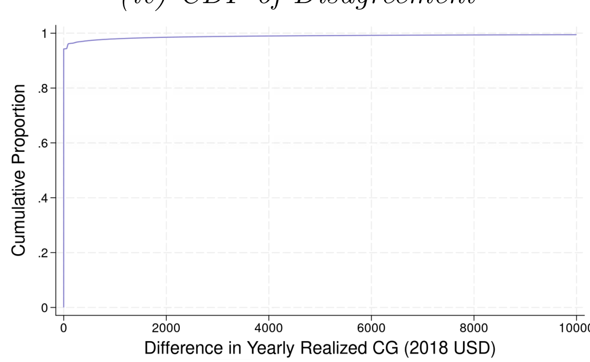


Panel B: Realized Capital Gains Comparison

(iii) Binned Scatter Comparison

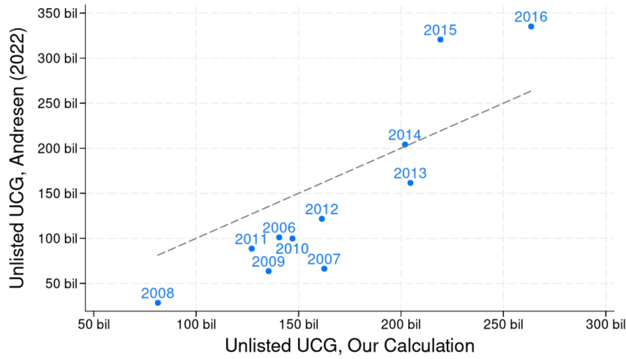


(iv) CDF of Disagreement



Note: Panel A compares our calculation of end-of-year shareholdings (at the individual-company-shareclass-year level) to stock ownership reported in the Shareholder Registry. This panel includes every shareholding of AS and ASA companies in our data from 2004-2018, whether or not we are able to match that shareholding into the Registry. 97% of all shareholdings match and agree exactly between the two data sources. Failures to find the shareholding (i.e., the year-person-company-shareclass record) in the Registry are coded as having zero Registry shares, and these failures to match account for the residual disagreement illustrated in Subfigure (ii). Panel B compares our calculation of individuals' yearly net realized capital gains in stock to non-real estate net realized capital gains reported in individuals' tax returns. This panel includes every person with an individual tax return from 2003-2004 and 2006-2018 (2005 is excluded due to our inability to separate out real estate realizations in the tax return data that year). 97% of all yearly realizations agree within rounding error. As depicted in Subfigure (iii), larger disagreements are associated with higher values in the tax returns than our own transaction-based calculation, consistent with the fact that the tax return-based measure includes realizations from non-real estate assets other than directly held stock.

Figure A2: Aggregate Unrealized Capital Gains (UCG) Comparison with Andresen (2022)



Note: This figure compares our series for the aggregate amount of unrealized capital gains in unlisted companies (x-axis, in billions of 2018 USD) to the same series from Andresen (2022) (y-axis) from 2006 to 2016. As a point of reference, the dashed line represents perfect agreement (y=x). The series from Andresen (2022) is taken from Figure 2(a) of that report, where it is reported (in 2021 kroner) as "skattepliktige latente eierinntekter" (taxable latent owner income), using the method of valuing unlisted stock with book-to-price multipliers. Values are then converted into 2018 dollars for comparison with our estimates. It is worth nothing that Andresen (2022) expresses substantial reservations about the quality of his data on basis. Given the data and methodological differences between our approaches (detailed in Appendix Section D.3), we consider the depicted level of agreement to be reasonable.

these data issues and the different procedure, his aggregate amounts of unrealized capital gains are broadly similar to our own estimates for unlisted firms (see Appendix Figure A2).

E Valuing Unlisted Firms in the Norwegian Data

Our preferred estimate of unlisted firm value largely follows the procedure of Smith et al. (2023) to compute a liquidity-adjusted, equal-weighted average of capitalized pro rata sales, assets, and EBITDA. Specifically, we apply a 10% liquidity discount to an equal-weighted average of three measures of the firm's capitalized value using (1) sales (2) assets and (3) EBITDA, respectively.

To capitalize each factor, we define valuation multiples for each NACE three-digit industry using Worldscope data on listed firms in the European Union, European Economic Area, and Switzerland.[58] We regress a ratio of the listed firm's equity value (i.e., market

unlisted shares is negative [...] it is unlikely that these negative basis values are real." (Andresen, 2022) (authors' own translation)

[58]Limiting ourselves to only Norwegian listed firms does not allow sufficient industry variation to estimate

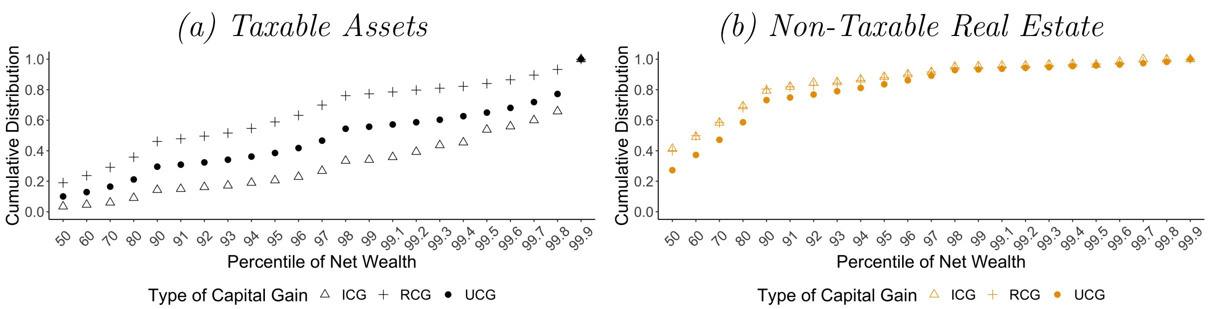
price times common shares outstanding) relative to the factor ($X \in \{sale, asset, ebitda\}$) on year-specific fixed effects for the firm's industry and country. We then employ the fitted values for Norway as our multiple for the given industry in the given year.

We assign unlisted firms in industries with insufficient data (less than 5 listed firms in the Worldscope data that year) a multiplier computed at a higher level in the industry classification (e.g., NACE 2, or the overall market average). We winsorize outlier multipliers at the 95th percentile. For firms that own other firms, we capitalize the firm value at the lowest level possible, and then sum over those valuations to value the parent firm. Since liquid financial assets are valued in the firm's book at market value, we value unlisted firms with substantial financial assets and little inherent value (holding companies) simply using the book values of assets minus liabilities.

F Additional Appendix Tables & Figures

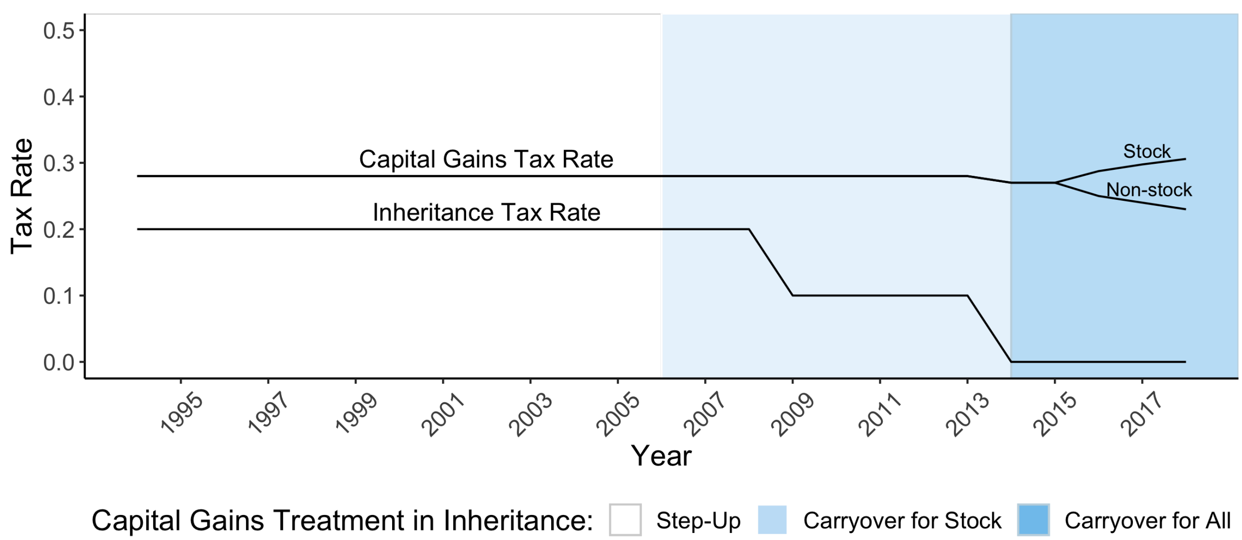
multipliers for all the industries of unlisted firms.

Figure A3: Comparison of Unrealized, Realized & Inherited Capital Gains, 2004



Note: This figure compares the cumulative distributions of unrealized capital gains, realized capital gains, and inherited capital gains, for household net wealth percentiles in 2004. Panel (a) displays these distributions for taxable assets (listed and unlisted stock and taxable real estate). Compared to realized capital gains, unrealized capital gains are more concentrated in wealthy households. Inherited capital gains are more concentrated in wealthy households than either realized or unrealized capital gains. Note that for comparability, these distributions all exclude possible capital gains from mutual funds. Panel (b) shows a corresponding figure for the distributions in non-taxable real estate. In contrast to the comparison for taxable assets, the distributions of realized, unrealized, and inherited capital gains in non-taxable real estate (the largest category of which is primary residences) are similarly distributed throughout the wealth distribution. For a presentation of the distribution of only unrealized capital gains (including mutual funds) or only inherited capital gains, see Figures 1 and 2, respectively.

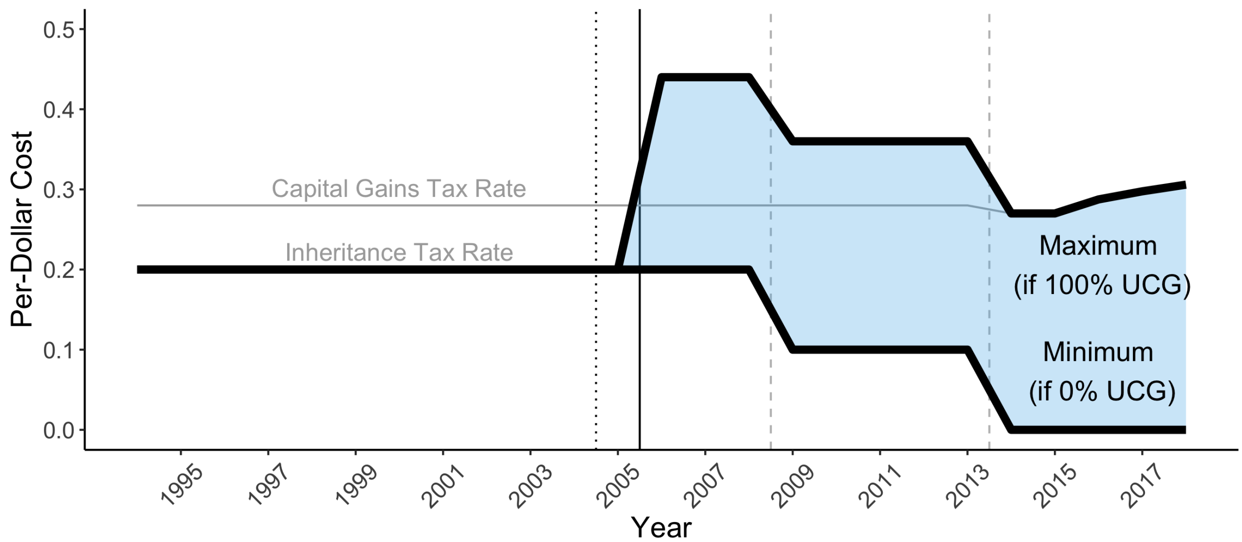
Figure A4: Capital Gains and Inheritance Policy in Norway, 1994–2018



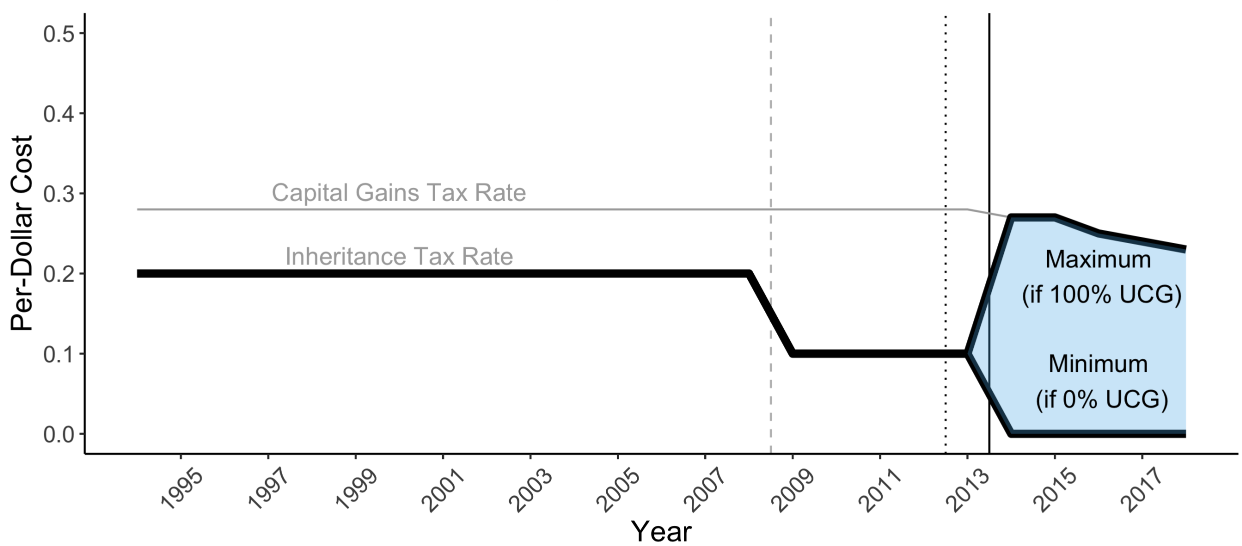
Note: This figure summarizes the changes to capital gains and inheritance taxation that took place during our study period. The statutory capital gains tax rate on all taxable assets was steady at 28% until 2014. In 2015, separate capital gains tax rates were introduced for stock and non-stock assets. Throughout our period, capital gains taxation in Norway has had a flat rate structure, although the rate was made to depend on the asset type (stock or non-stock) in later years. In contrast, inheritance tax rates (prior to their abolition) were progressive in the cumulative amount of the inheritance received, and depended on the relationship between giver and recipient. This graph depicts the top statutory inheritance tax rate for closely related family members, which fell from 20% to 10% in 2009 and to 0% with the abolition of the inheritance tax in 2014. See Appendix Table A1 for more information on inheritance tax rates and brackets. The shading of the graph depicts changes to the treatment of inherited capital gains. Prior to 2006, all gifted and inherited assets benefited from stepped-up basis. From 2006, gifted and inherited stock with capital gains were switched to a system of carryover basis. From 2014, all gifted and inherited assets carried their basis over to the recipient, unless the asset would not have been subject to capital gains taxation in the hands of the original owner.

Figure A5: Per-Dollar Inheritance Costs, by Year & Asset Class

(a) Stock

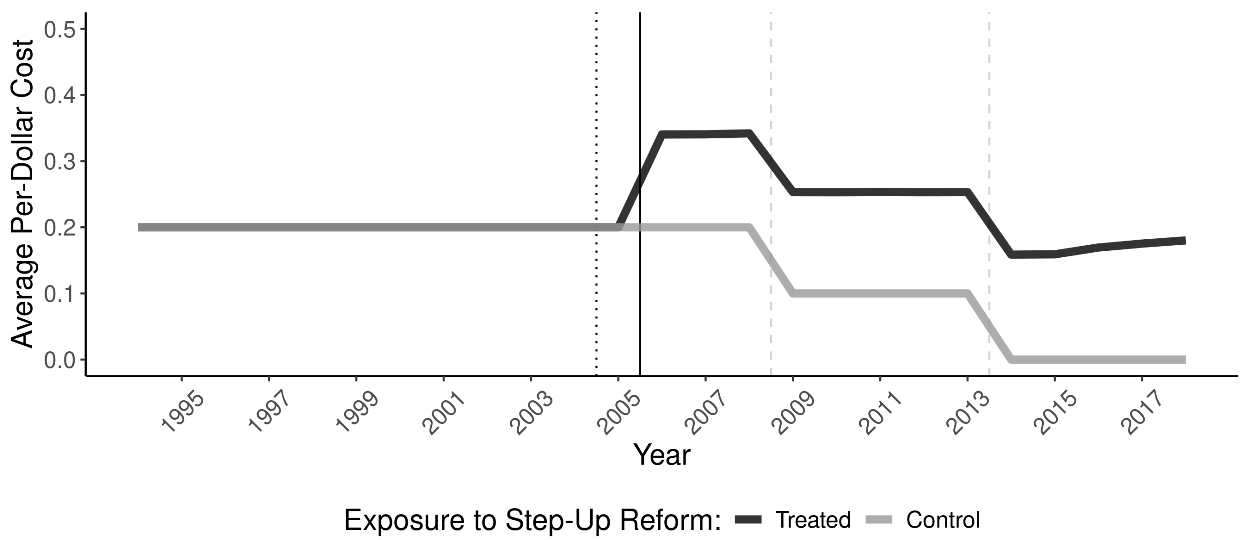


(b) Non-Stock



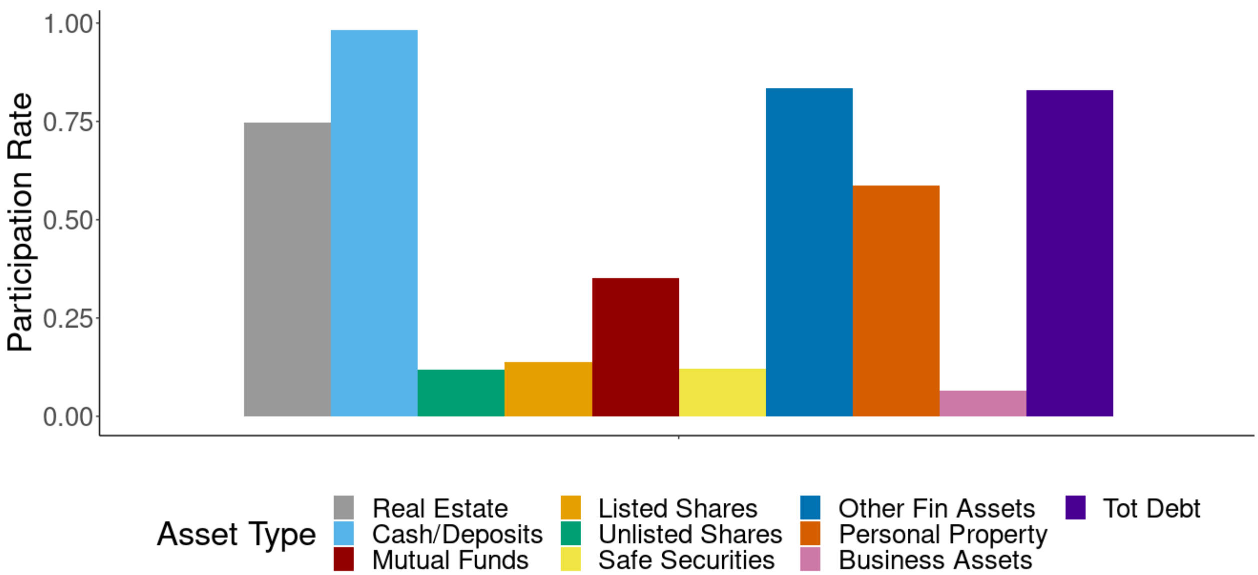
Note: This figure depicts the range of per-dollar costs of inheritance $c_I = \tau_I + \theta\tau_c\gamma$ for assets of a given class (stock or non-stock) from 1994–2018. Prior to 2006, Norway had a stepped-up basis regime ($\theta = 0$), so all assets had a inheritance cost equal to the statutory inheritance tax rate (τ_i =20% in the highest bracket, for inheritance between relatives). Carryover basis ($\theta = 1$) is introduced for stock in 2006 (Panel a) and other assets in 2014 (Panel b), which widens the range of effective tax rates on inherited assets. When carryover basis is in place, assets with 0% unrealized capital gains still face only the statutory inheritance tax, but assets with positive unrealized capital gains face both the statutory inheritance tax and a capital gains tax. At the maximum, assets with 100% unrealized capital gains in a carryover basis regime have an effective tax rate equal to the sum of the capital gains and inheritance tax rates. The dotted black lines (before 2005 or 2013) indicate the announcement of step-up's impending removal for the given asset class. The solid black lines (before 2006 or 2014) indicates the implementation of step-up's removal for that asset class. The dashed grey lines (before 2009 and 2014) indicate reductions to the statutory inheritance tax rate.

Figure A6: Average Per-Dollar Inheritance Costs (c_I), for Treated and Control Group



Note: As an extension of Table 2 Panel (a), this figure depicts the empirical average of our measure of the "per-dollar cost of inheritance" from 1994 to 2018 for the treatment and control groups in our analysis of the removal of step-up for stock. The per-dollar cost of inheritance is defined as $c_I = \tau_I + \theta_t \tau_c \gamma$ in Section 4.2. Before 2006, step-up is in place ($\theta = 0$), so the total cost is simply the statutory inheritance tax rate (τ_i). Once step-up is removed ($\theta = 1$), costs begin to depend on the amount of appreciation in each individual portfolio. The dotted black line before 2005 indicates the announcement of step-up's impending removal for stock. The solid black line before 2006 indicates the implementation of step-up's removal for stock. The dashed grey line before 2009 indicates a fall (by half) in the statutory inheritance tax rate. The dashed grey line before 2014 indicates the full removal of the inheritance tax, and the beginning of a period of increasing capital gains tax rates.

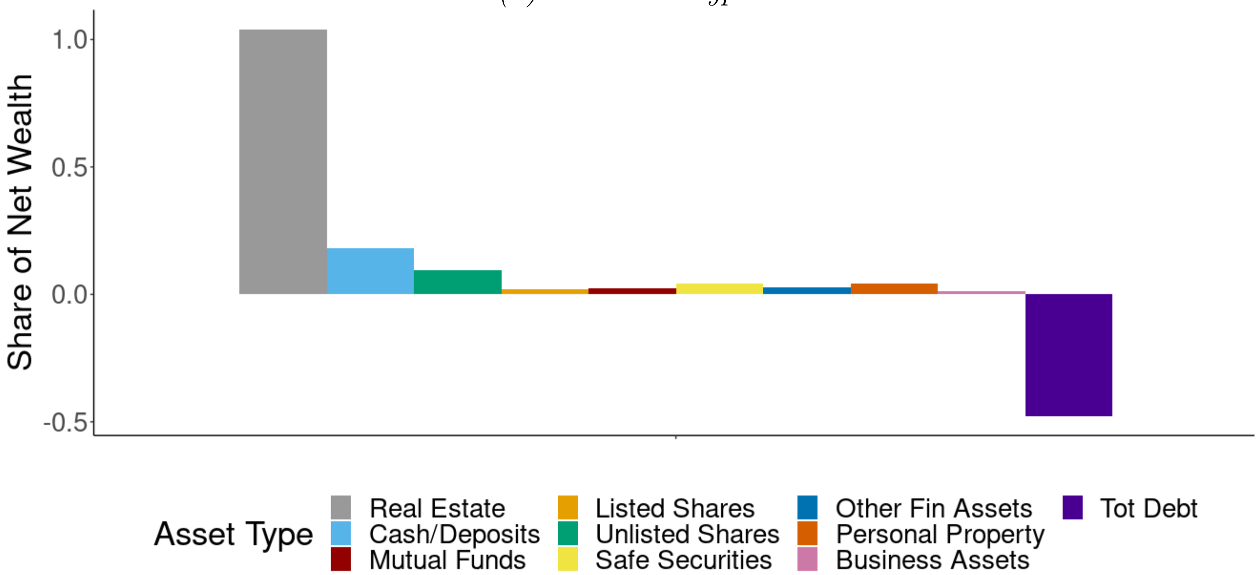
Figure A7: Ownership Rates by Asset Category in 2004



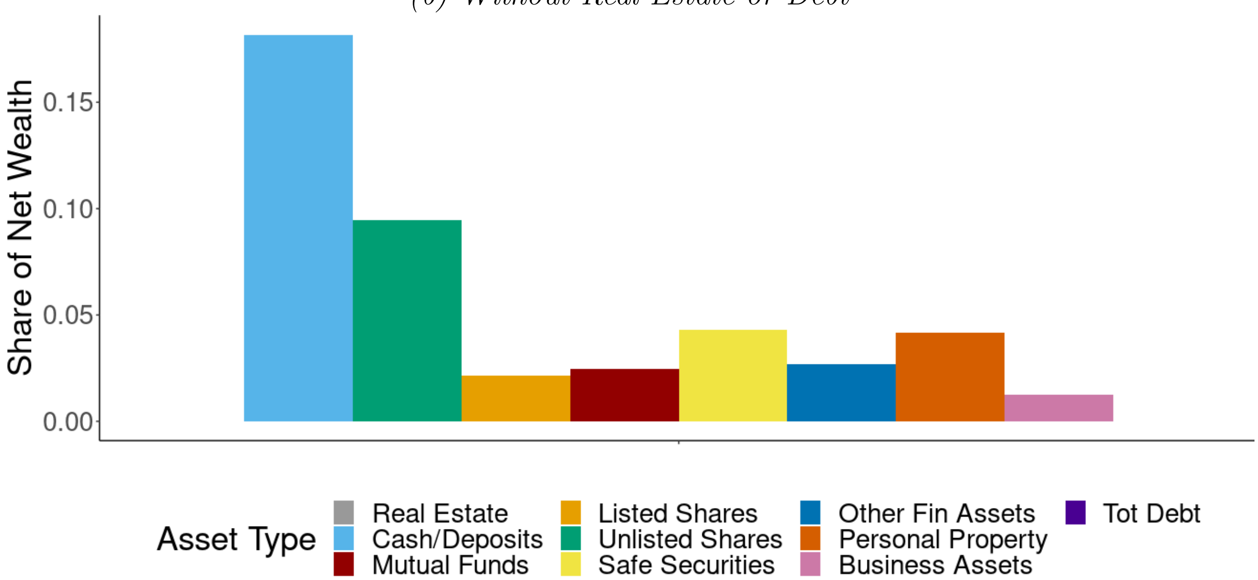
Note: This figure depicts the percentage of Norwegian households that hold each asset type at the end of 2004. 14% of Norwegian households owned listed stock and 12% of Norwegian households owned unlisted stock in 2004.

Figure A8: Average Portfolio Shares in 2004

(a) All Asset Types



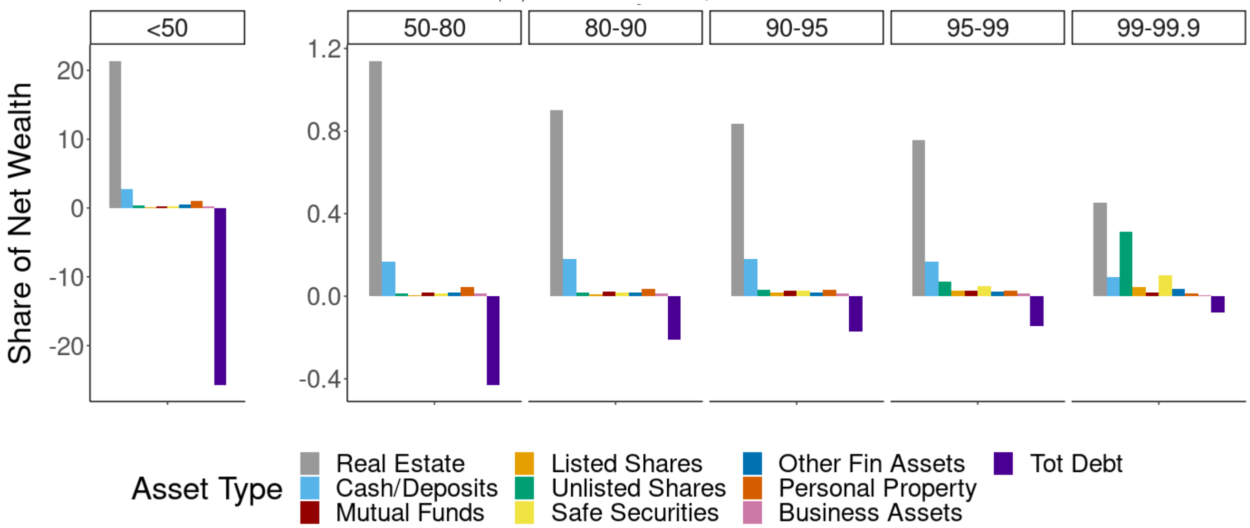
(b) Without Real Estate or Debt



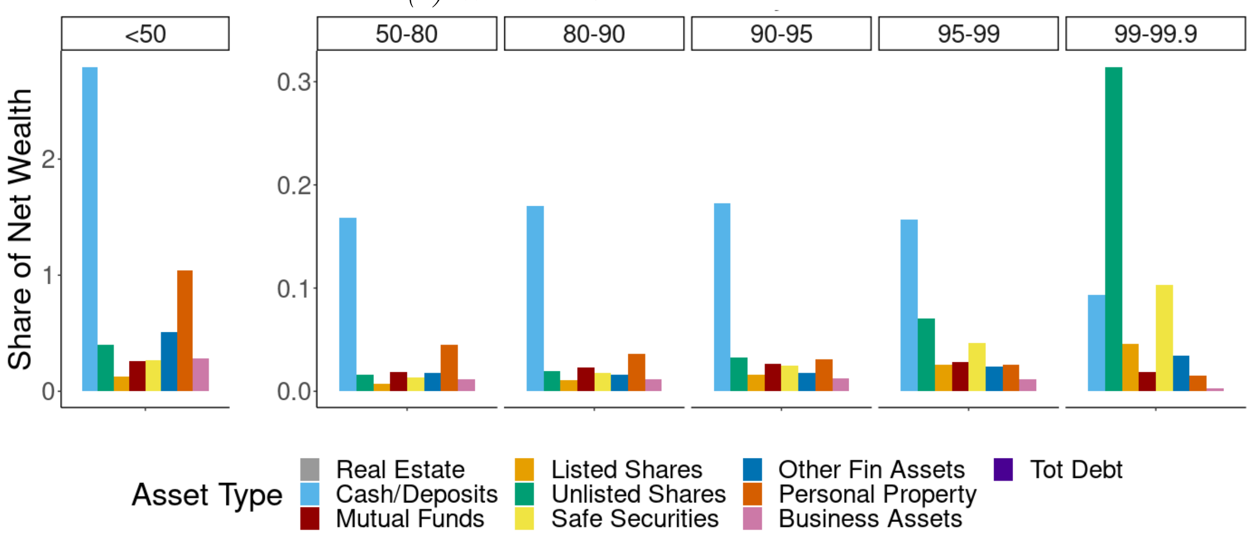
Note: This figure depicts portfolio shares for the average Norwegian household at the end of 2004. Panel (a) presents all asset types. Panel (b) is a zoomed in version for those asset types other than real estate and debt.

Figure A9: Average Portfolio Shares in 2004, by Net Worth Percentile

(a) All Asset Types



(b) Without Real Estate or Debt



Note: This figure depicts portfolio shares for Norwegian household broken out by percentile of net worth at the end of 2004. Panel (a) presents all asset types. Panel (b) is a zoomed in version for those asset types other than real estate and debt. Note that averages below the 50th percentile include people with negative net worth (lots of debt), implying portfolio shares greater than 1. Unlisted stock is a disproportionately important part of very rich people's portfolios – people between the 99th and 99.99th percentile of net wealth in 2004 had an average holding of \$1.3 million (31% of net wealth).

Table A1: Inheritance Tax Rates, 1985-2013

Years	Amount Received (NOK)	Tax Rate		
		Spouses	Close Relatives	Other Recipients
1985-1998	0 - 100,000	0	0	0
1985-1998	100,000 - 400,000	0	0.08	0.10
1985-1998	400,000 +	0	0.20	0.30
1999-2002	0 - 200,000	0	0	0
1999-2002	200,000 - 500,000	0	0.08	0.10
1999-2002	500,000 +	0	0.20	0.30
2003-2008	0 - 250,000	0	0	0
2003-2008	250,000 - 550,000	0	0.08	0.10
2003-2008	550,000 +	0	0.20	0.30
2009-2013	0 - 470,000	0	0	0
2009-2013	470,000 - 800,000	0	0.06	0.08
2009-2013	800,000 +	0	0.10	0.15

Note: This table depicts the Norwegian inheritance tax scheme from 1985 until the inheritance tax was abolished in 2014. Inheritance tax was levied without distinction between *inter vivos* gifts and bequests at the time of death. The rate structure depended on the cumulative amount of the gift/inheritance received from the individual giver, and the relationship between the giver and the recipient. The definition of close relatives includes parents and children of the giver. Transfers between spouses were not taxed. Transfers between individuals who are not close relatives were subject to a higher top rate in every year. In 2014, the gift and inheritance tax was abolished, so all rates for subsequent years are set to 0%.